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FURTHER MATHEMATICS • UPPER SIXTH

UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 9

SIMPLE AND DAMPED HARMONIC MOTION

SUGGESTED TIME: 3 PERIODS

IN THIS CHAPTER

This chapter studies simple harmonic motion (SHM) and damped harmonic motion through their defining differential equations, their solutions, and their energy properties. It closes with modelling real oscillating systems — springs, pendulums, vehicle suspensions — in the spirit of GCE Advanced Level problems.

- By the end of this chapter I can recognise a physical situation that leads to simple harmonic motion and derive the equation $\ddot{x} = -\omega^2 x$ from Newton's second law.
- By the end of this chapter I can solve the SHM equation and write the solution in the forms $x = a\sin(\omega t + \varepsilon)$ and $x = a\cos(\omega t + \varphi)$, choosing the form adapted to the initial conditions.
- By the end of this chapter I can determine the amplitude, period, frequency and phase of an SHM and compute the velocity and acceleration at any point of the motion.
- By the end of this chapter I can use the relation $v^2 = \omega^2(a^2 - x^2)$ to link speed and position, and analyse the energy of an oscillator.
- By the end of this chapter I can set up the damped harmonic motion equation $m\ddot{x} = -kx - \lambda\dot{x}$ and identify the damping coefficient.
- By the end of this chapter I can solve the damped equation using the auxiliary (characteristic) equation and distinguish light, critical and heavy damping.

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THIS SIGNED COURSE CONTAINS

COURSE — complete, illustrated, step by step

METHODS — and worked examples

EXERCISES — exam style + solutions

SUMMARY SHEET — the essentials to remember

CONTENTS

1. Simple Harmonic Motion: Definition and Equation
2. Further Results on SHM: Energy, Pendulums and Composite Problems
3. Damped Harmonic Motion
4. Applications and GCE Problem-Solving
5. Integration activity
6. Methods and worked examples
7. Exercises
8. Solutions to the exercises
9. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can recognise a physical situation that leads to simple harmonic motion and derive the equation $\ddot{x} = -\omega^2 x$ from Newton's second law.
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- By the end of this chapter I can solve the damped equation using the auxiliary (characteristic) equation and distinguish light, critical and heavy damping.
- By the end of this chapter I can sketch and interpret the displacement–time graphs of under-damped, critically damped and over-damped motions.
- By the end of this chapter I can model concrete systems (mass–spring, simple pendulum for small angles, vehicle suspension) and compute their periods or damping behaviour.
- By the end of this chapter I can solve full GCE-style problems combining derivation of the equation of motion, solution, and interpretation of results.

PREREQUISITES

- Second-order linear differential equations with constant coefficients and the auxiliary equation (Lower Sixth / Upper Sixth pure mathematics)
- Newton's second law of motion and resolution of forces (Lower Sixth mechanics)
- Hooke's law for springs and elastic strings: $T = \lambda x / l$ (Lower Sixth mechanics)
- Trigonometric functions, their derivatives, and the $R\sin(\omega t + \varepsilon)$ compound-angle form
- Kinematics: velocity and acceleration as first and second derivatives of displacement

1 SIMPLE HARMONIC MOTION: DEFINITION AND EQUATION

Every evening in Bamenda, a trader hangs her spring scale and watches the pointer bounce up and down before settling on the weight of a bag of beans. On the road to Douala, passengers feel their bus bounce after each pothole, yet the oscillations die out quickly thanks to the shock absorbers. Why does the pointer oscillate, why does the motion gradually fade, and how can we predict the period of these vibrations? This chapter develops the mathematics — differential equations of the form $\ddot{x} = -\omega^2 x$ and $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ — that answer these questions and appear every year in the GCE Further Mathematics paper.

1.1 What is Simple Harmonic Motion?

DEFINITION **Simple Harmonic Motion (SHM)**

A particle P describes simple harmonic motion about a fixed centre O if its acceleration is always directed towards O and is proportional to its displacement from O. If x is the displacement from O at time t, then

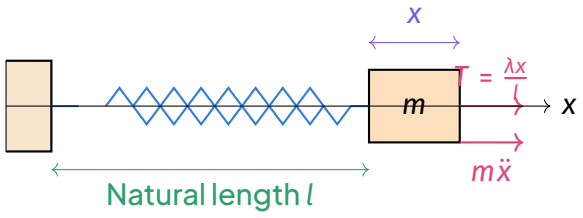
$$\ddot{x} = -\omega^2 x$$

where $\omega > 0$ is a constant called the **angular frequency**.

The negative sign is crucial: it tells us that the acceleration is restoring — it always points back towards the centre O. The constant ω^2 is the constant of proportionality between the magnitude of the acceleration and the displacement.

1.2 Derivation from a Horizontal Mass–Spring System

Consider a particle of mass m attached to one end of a light elastic spring of natural length l and modulus of elasticity λ. The other end of the spring is fixed at a point A on a smooth horizontal table. Let O be the point where the spring is unstretched (natural length). At time t, let the displacement of P from O be x (positive in the direction A to P). The extension of the spring is then x.



Mass–spring system on a smooth horizontal table. The tension T acts as the restoring force.

By Hooke's law, the tension in the spring is $T = \dfrac{\lambda x}{l}$. This tension acts towards A (i.e. in the nega-

tive x direction). Applying Newton's second law in the positive x direction:

$$-m\ddot{x} = T = \frac{\lambda x}{l} \quad \Longrightarrow \quad \ddot{x} = -\frac{\lambda}{ml}x.$$

This is exactly the SHM equation $\ddot{x} = -\omega^2 x$ with

$$\omega^2 = \frac{\lambda}{ml}.$$

DID YOU KNOW? **Modulus of Elasticity**

The modulus λ (measured in newtons) is the force required to double the length of the spring. For a spring of natural length l, the tension for an extension x is $T = \frac{\lambda x}{l}$. This is Hooke's law in the form used in the GCE syllabus. Do not confuse λ with the spring constant $k = \lambda/l$ (stiffness, in N/m).

1.3 Equilibrium Position and Change of Variable

In many problems the spring is not at its natural length when the system is at rest. For example, a vertical spring stretches under the weight of the mass. The **equilibrium position** E is where the net force is zero. If we measure displacement X from E (i.e. $X = x - e$ where e is the equilibrium extension), then the equation of motion becomes $\ddot{X} = -\omega^2 X$, proving SHM about E.

METHOD **Proving a Motion is SHM**

1. Identify the forces acting on the particle.
2. Find the equilibrium position E (where resultant force = 0).
3. Let X be the displacement from E (choose a positive direction).
4. Apply $F = ma$ in the positive direction, expressing all forces in terms of X.
5. Show that the equation reduces to $\ddot{X} = -\omega^2 X$ and read off ω.

WARNING **Amplitude vs. Initial Displacement**

The **amplitude** a is the maximum displacement from the centre of oscillation (equilibrium position), not from the spring's natural length. Always locate the equilibrium position first! A common mistake is to take the initial extension from the natural length as the amplitude.

1.4 General Solution and Initial Conditions

The differential equation $\ddot{x} = -\omega^2 x$ has general solution

$$x = a\sin(\omega t + \varepsilon)$$

where $a \geq 0$ is the **amplitude** and ε is the **phase constant** (or initial phase). Equivalently,

$$x = A\cos\omega t + B\sin\omega t \quad \text{with} \quad a = \sqrt{A^2 + B^2},\ \tan\varepsilon = \frac{A}{B}.$$

The constants are determined by the initial conditions $x(0)$ and $\dot{x}(0)$:

$$x(0) = a\sin\varepsilon, \qquad \dot{x}(0) = a\omega\cos\varepsilon.$$

KEY POINTS **GCE Shortcut**

- If the particle is released from rest at displacement $x = a$ (maximum displacement), then $\varepsilon = \frac{\pi}{2}$ and $x = a\cos\omega t$.
- If the particle is projected from the centre $x = 0$ with speed v_0, then $\varepsilon = 0$ and $x = \frac{v_0}{\omega}\sin\omega t$.

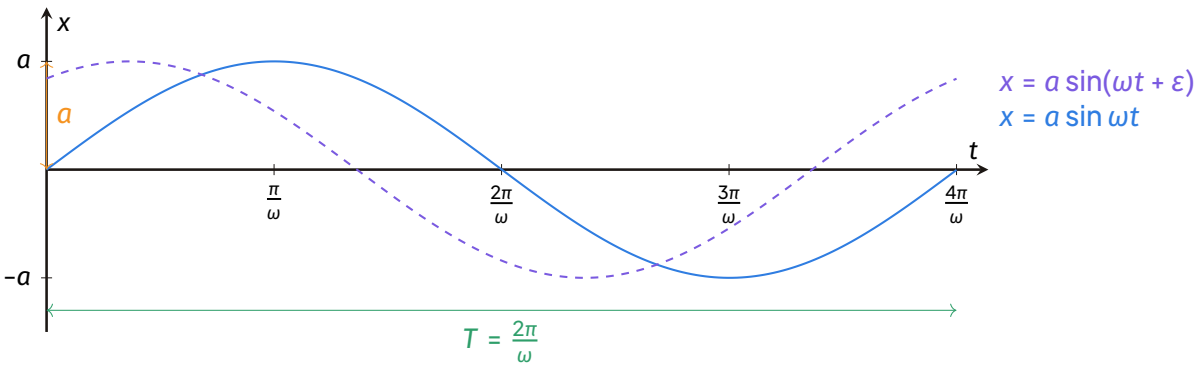
This avoids solving for ε explicitly.

1.5 Key Vocabulary and Quantities

PROPERTY **Parameters of SHM**

For a particle executing SHM $x = a\sin(\omega t + \varepsilon)$:

- Amplitude a: maximum displacement from the centre.
- Angular frequency ω: $\omega = \sqrt{\frac{\lambda}{ml}}$ (rad/s).
- Period $T = \frac{2\pi}{\omega}$: time for one complete oscillation.
- Frequency $f = \frac{1}{T} = \frac{\omega}{2\pi}$: number of oscillations per second (Hz).
- Phase at time t: $\phi = \omega t + \varepsilon$.



Graph of $x = a\sin(\omega t+\varepsilon)$ showing amplitude a and period T. The dashed curve shows a phase shift ε.

1.6 Velocity and Acceleration in SHM

Differentiating $x = a\sin(\omega t + \varepsilon)$:

$$\dot{x} = a\omega\cos(\omega t + \varepsilon), \qquad \ddot{x} = -a\omega^2\sin(\omega t + \varepsilon) = -\omega^2 x.$$

PROPERTY **Maximum Values**

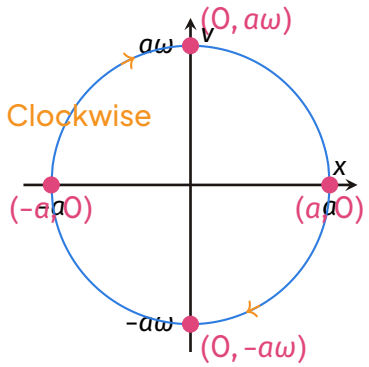
- Maximum speed: $v_{\max} = a\omega$ (occurs at the centre $x = 0$).
- Maximum acceleration: $|\ddot{x}|_{\max} = a\omega^2$ (occurs at the extremes $x = \pm a$).

1.7 The Speed–Displacement Relation

Eliminating t between $x = a\sin(\omega t + \varepsilon)$ and $\dot{x} = a\omega\cos(\omega t + \varepsilon)$ using $\sin^2\theta + \cos^2\theta = 1$ gives the fundamental relation:

$$v^2 = \omega^2(a^2 - x^2) \quad \text{where } v = \dot{x}.$$

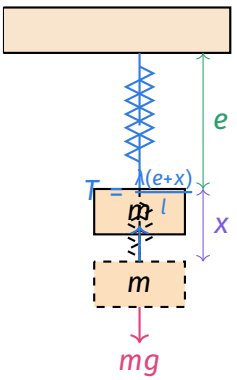
This tells us that the speed depends only on the displacement, not on the direction of motion.



The ellipse $v^2 = \omega^2(a^2 - x^2)$ in the phase plane. The motion proceeds clockwise.

1.8 Vertical Oscillations of a Spring

A light spring of natural length l and modulus λ hangs vertically from a fixed point. A mass m is attached to the lower end. At equilibrium, the extension is e where $mg = \dfrac{\lambda e}{l}$. Let x be the displacement **downwards** from the equilibrium position. The total extension is $e + x$.



Vertical mass–spring system. The equilibrium extension is e; x is measured downwards from equilibrium.

Applying Newton's second law downwards:

$$m\ddot{x} = mg - T = mg - \frac{\lambda(e + x)}{l}.$$

But $mg = \dfrac{\lambda e}{l}$, so

$$m\ddot{x} = -\frac{\lambda x}{l} \quad \Longrightarrow \quad \ddot{x} = -\frac{\lambda}{ml}x.$$

This is SHM about the equilibrium position with the **same** angular frequency $\omega^2 = \dfrac{\lambda}{ml}$ as the horizontal case. The period is independent of g!

KEY POINTS **Key Results for Vertical Spring**

- Equilibrium extension: $e = \dfrac{mgl}{\lambda}$.
- Equation of motion about equilibrium: $\ddot{x} = -\dfrac{\lambda}{ml}x$.
- Angular frequency: $\omega = \sqrt{\dfrac{\lambda}{ml}}$.
- Period: $T = 2\pi\sqrt{\dfrac{ml}{\lambda}}$.

1.9 Worked Examples

WORKED EXAMPLE 1 **Horizontal Spring on a Smooth Table**

A particle P of mass 0.5 kg is attached to one end of a light elastic spring of natural length 1.2 m and modulus 20 N. The other end is fixed at A on a smooth horizontal table. P is held at rest with $AP = 1.5$ m and then released.

1. Show that P moves with SHM and find the period.
2. Find the maximum speed of P.
3. Find the speed of P when $AP = 1.3$ m.

SOLUTION

Solution.

1. The equilibrium position is at natural length $l = 1.2$ m (no other horizontal forces). Let x be the displacement from equilibrium (positive towards A). Extension $= x$. Tension $T = \dfrac{\lambda x}{l} = \dfrac{20x}{1.2} = \dfrac{50}{3}x$ N. Equation of motion: $-T = m\ddot{x} \Rightarrow -\dfrac{50}{3}x = 0.5\ddot{x} \Rightarrow \ddot{x} = -\dfrac{100}{3}x$. This is SHM with $\omega^2 = \dfrac{100}{3}$, so $\omega = \dfrac{10}{\sqrt{3}}$ rad/s. Period $T = \dfrac{2\pi}{\omega} = 2\pi\dfrac{\sqrt{3}}{10} = \dfrac{\pi\sqrt{3}}{5} \approx 1.088$ s.
2. Amplitude a = initial displacement from equilibrium $= 1.5 - 1.2 = 0.3$ m. Maximum speed $v_{\max} = a\omega = 0.3 \times \dfrac{10}{\sqrt{3}} = \dfrac{3}{\sqrt{3}} = \sqrt{3} \approx 1.732$ m/s.
3. When $AP = 1.3$ m, displacement from equilibrium $x = 1.3 - 1.2 = 0.1$ m. Using $v^2 = \omega^2(a^2 - x^2) = \dfrac{100}{3}(0.3^2 - 0.1^2) = \dfrac{100}{3}(0.09 - 0.01) = \dfrac{100}{3} \times 0.08 = \dfrac{8}{3}$. Speed $v = \sqrt{\dfrac{8}{3}} = \dfrac{2\sqrt{2}}{\sqrt{3}} \approx 1.633$ m/s.

WORKED EXAMPLE 2 **Vertical Spring — Finding Amplitude and Period**

A light elastic string of natural length 0.8 m and modulus 12 N has one end fixed at A. A particle of mass 0.4 kg is attached to the other end and hangs in equilibrium. The particle is pulled down a further 0.2 m and released from rest.

1. Find the equilibrium extension.
2. Show that the subsequent motion is SHM and find its period.
3. Find the maximum speed of the particle.

SOLUTION

Solution.

1. At equilibrium, $mg = \frac{\lambda e}{l} \Rightarrow 0.4 \times 9.8 = \frac{12e}{0.8} \Rightarrow 3.92 = 15e \Rightarrow e = \frac{3.92}{15} = 0.2613\,\text{m}$ (approx).
2. Let x be displacement downwards from equilibrium. Total extension $= e + x$. Tension $T = \frac{\lambda(e+x)}{l}$. Equation downwards: $m\ddot{x} = mg - T = mg - \frac{\lambda(e+x)}{l} = mg - \frac{\lambda e}{l} - \frac{\lambda x}{l} = -\frac{\lambda x}{l}$. So $\ddot{x} = -\frac{\lambda}{ml}x = -\frac{12}{0.4\times0.8}x = -37.5x$. SHM with $\omega^2 = 37.5$, $\omega = \sqrt{37.5} = \frac{5\sqrt{6}}{2} \approx 6.124\,\text{rad/s}$. Period $T = \frac{2\pi}{\omega} = \frac{4\pi}{5\sqrt{6}} \approx 1.026\,\text{s}$.
3. Amplitude $a = 0.2\,\text{m}$ (the additional displacement from equilibrium). Maximum speed $v_{\max} = a\omega = 0.2 \times \sqrt{37.5} = \frac{\sqrt{37.5}}{5} \approx 1.225\,\text{m/s}$.

1.10 Exercises

EXERCISE 1 **Exercise 1**

A particle of mass 2 kg is attached to one end of a light elastic spring of natural length 0.5 m and modulus 50 N. The other end is fixed on a smooth horizontal table. The particle is pulled until the spring length is 0.8 m and released from rest.

1. Prove that the motion is SHM and find ω.
2. Find the period and amplitude.
3. Calculate the maximum acceleration.
4. Find the speed when the spring length is 0.6 m.

EXERCISE 2 **Exercise 2**

A light elastic string of natural length 1 m and modulus 20 N hangs vertically from a fixed

point. A mass of 0.5 kg is attached to the lower end. The mass is pulled down 0.15 m below the equilibrium position and released from rest.

1. Find the equilibrium extension.
2. Show that the motion is SHM and find its period.
3. Find the greatest speed of the mass.
4. Find the time taken for the mass to first reach a point 0.05 m above the equilibrium position.

EXERCISE 3 **Exercise 3**

A particle P moves with SHM about centre O with period $\frac{2\pi}{3}$ s. When P is at a distance 0.4 m from O, its speed is 1.2 m/s. Find the amplitude of the motion and the maximum speed.

SOLUTION **Exercise 1**

1. Equilibrium at natural length $l = 0.5$. Let x be extension. $T = \frac{50x}{0.5} = 100x$. $-T = 2\ddot{x} \Rightarrow \ddot{x} = -50x$. $\omega^2 = 50$, $\omega = 5\sqrt{2}$.
2. $T = \frac{2\pi}{5\sqrt{2}} = \frac{\pi\sqrt{2}}{5}$ s. Amplitude $a = 0.8 - 0.5 = 0.3$ m.
3. $|\ddot{x}|_{\max} = a\omega^2 = 0.3 \times 50 = 15\ \text{m/s}^2$.
4. Spring length $0.6 \Rightarrow x = 0.1$. $v^2 = 50(0.3^2 - 0.1^2) = 50 \times 0.08 = 4 \Rightarrow v = 2$ m/s.

SOLUTION **Exercise 2**

1. $mg = \frac{\lambda e}{l} \Rightarrow 0.5 \times 9.8 = 20e \Rightarrow e = 0.245$ m.
2. $\ddot{x} = -\frac{\lambda}{ml}x = -\frac{20}{0.5\times1}x = -40x$. $\omega = \sqrt{40} = 2\sqrt{10}$. $T = \frac{2\pi}{2\sqrt{10}} = \frac{\pi}{\sqrt{10}}$ s.
3. $a = 0.15$. $v_{\max} = a\omega = 0.15 \times 2\sqrt{10} = 0.3\sqrt{10} \approx 0.949$ m/s.
4. $x = a\cos\omega t$ (released from rest at $x = a$). $0.05 = 0.15\cos(2\sqrt{10}t) \Rightarrow \cos(2\sqrt{10}t) = 1/3 \Rightarrow 2\sqrt{10}t = \arccos(1/3) \approx 1.231 \Rightarrow t \approx 0.194$ s.

SOLUTION **Exercise 3**

$T = \frac{2\pi}{3} \Rightarrow \omega = 3$. Given $x = 0.4$, $v = 1.2$. $v^2 = \omega^2(a^2 - x^2) \Rightarrow 1.44 = 9(a^2 - 0.16) \Rightarrow a^2 - 0.16 = 0.16 \Rightarrow a^2 = 0.32 \Rightarrow a = 0.4\sqrt{2} \approx 0.566\,\text{m}$. $v_{\max} = a\omega = 1.2\sqrt{2} \approx 1.697\,\text{m/s}$.

2 FURTHER RESULTS ON SHM: ENERGY, PENDULUMS AND COMPOSITE PROBLEMS

In the previous section we established the definition and the differential equation of simple harmonic motion (SHM): $\ddot{x} = -\omega^2 x$. We also learned how to write the displacement as $x = a\cos(\omega t + \varepsilon)$ or $x = a\sin(\omega t + \varepsilon)$, and how to obtain the velocity $\dot{x} = \pm\omega\sqrt{a^2 - x^2}$ and the acceleration. We now deepen our understanding by studying energy in SHM, the classic example of the simple pendulum, systems with two springs or strings, motions in which the SHM is interrupted (for example when a string goes slack), and the important technique of calculating the time taken to move between two points. These are precisely the tools that turn the abstract equation into a powerful model for GCE examination problems.

2.1 Energy in Simple Harmonic Motion

Consider a particle of mass m attached to a light elastic string or spring of natural length l and modulus of elasticity λ, performing SHM on a smooth horizontal table. The only force doing work is the elastic tension (or thrust), which is conservative. Therefore the total mechanical energy is conserved.

The kinetic energy is $KE = \frac{1}{2}m\dot{x}^2$. The elastic potential energy stored in a string or spring stretched by x is $PE = \frac{\lambda x^2}{2l}$. Writing Hooke's law as $T = \frac{\lambda x}{l} = kx$ with stiffness $k = \lambda/l$, this becomes $PE = \frac{1}{2}kx^2$.

From the SHM equation $\ddot{x} = -\frac{\lambda}{ml}x$ we read off $\omega^2 = \frac{\lambda}{ml}$ (equivalently $\omega^2 = \frac{k}{m}$). Hence $\lambda = m\omega^2 l$ and the potential energy measured from the centre of oscillation is $PE = \frac{1}{2}m\omega^2 x^2$.

PROPERTY **Conservation of Energy in SHM**

In SHM with amplitude a and angular frequency ω, the total mechanical energy E is constant and is given by

$$E = \frac{1}{2}m\omega^2 a^2.$$

At displacement x from the centre,

$$KE = \frac{1}{2}m\omega^2(a^2 - x^2), \qquad PE = \frac{1}{2}m\omega^2 x^2.$$

Kinetic energy is maximum at the centre ($x = 0$) and zero at the extremes ($x = \pm a$); potential energy is zero at the centre and maximum at the extremes.

Proof (examinable as a "show that"). Take $x = a\cos(\omega t + \varepsilon)$, so $\dot{x} = -a\omega\sin(\omega t + \varepsilon)$. Using $\dot{x}^2 = \omega^2(a^2 - x^2)$,

$$KE = \frac{1}{2}m\omega^2(a^2 - x^2), \qquad PE = \frac{1}{2}m\omega^2 x^2.$$

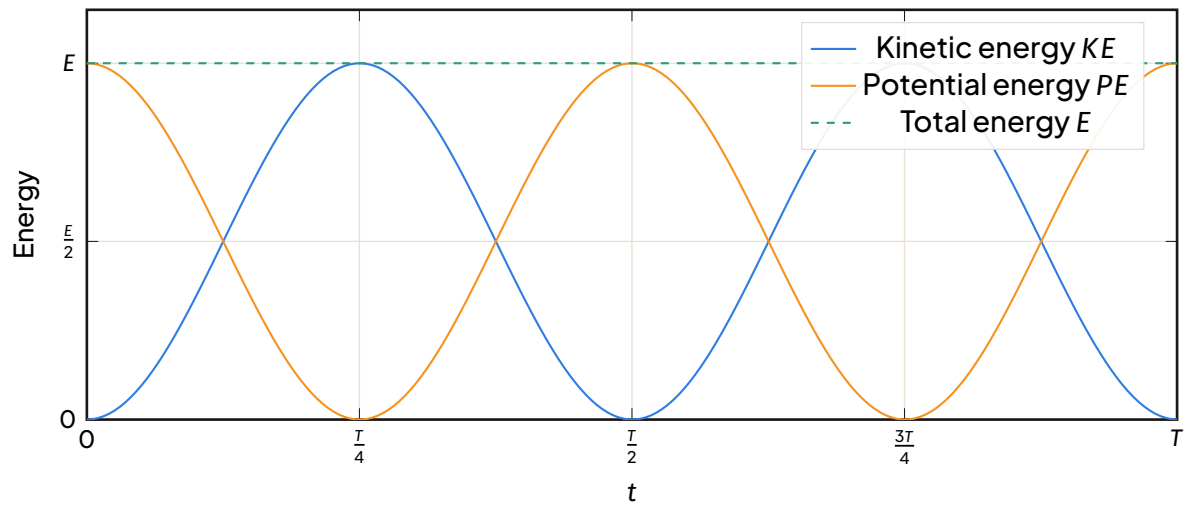
Adding,

$$E = KE + PE = \frac{1}{2}m\omega^2(a^2 - x^2) + \frac{1}{2}m\omega^2 x^2 = \frac{1}{2}m\omega^2 a^2,$$

which is independent of x and of t: the total energy is constant. Equivalently, substituting the trigonometric forms gives $E = \frac{1}{2}m\omega^2 a^2(\sin^2(\omega t + \varepsilon) + \cos^2(\omega t + \varepsilon)) = \frac{1}{2}m\omega^2 a^2$. ■

KEY POINTS **Energy Summary**

- Total energy $E = \frac{1}{2}m\omega^2 a^2$ (constant).
- $KE = \frac{1}{2}m\omega^2(a^2 - x^2)$; $PE = \frac{1}{2}m\omega^2 x^2$.
- KE and PE each vary sinusoidally with time at twice the frequency of the motion (period $T/2$), because they involve $\sin^2$ and $\cos^2$.
- The average kinetic energy over one period equals the average potential energy, each being half the total: $\langle KE\rangle = \langle PE\rangle = \frac{1}{4}m\omega^2 a^2$.



Variation of kinetic, potential and total energy with time over one period T (particle released from $x = +a$ at $t = 0$). Note that KE and PE oscillate at twice the frequency of the motion, while their sum stays constant.

WORKED EXAMPLE 3 **Energy in a vertical string**

A particle of mass 0.5 kg hangs in equilibrium from a light elastic string of natural length 1 m and modulus 20 N. The particle is pulled down a further 0.2 m and released from rest. Find (a) the angular frequency ω, (b) the total energy of the motion, (c) the maximum speed. (Take $g = 9.8\ \text{m s}^{-2}$.)

SOLUTION

(a) At equilibrium the tension balances the weight: $mg = \dfrac{\lambda e}{l}$, giving $e = \dfrac{mgl}{\lambda} = \dfrac{0.5 \times 9.8 \times 1}{20} = 0.245$ m. For vertical oscillations about the equilibrium position the equation of motion reduces to $\ddot{x} = -\dfrac{\lambda}{ml}x$ (the weight cancels the equilibrium tension), so

$$\omega^2 = \frac{\lambda}{ml} = \frac{20}{0.5 \times 1} = 40 \quad \Rightarrow \quad \omega = 2\sqrt{10}\ \text{rad s}^{-1}.$$

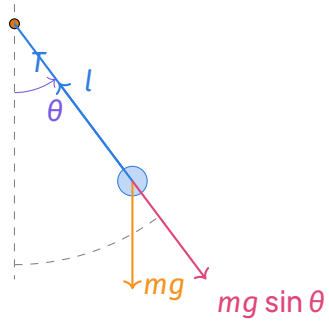
(b) The amplitude is $a = 0.2$ m (released from rest 0.2 m below equilibrium). Hence

$$E = \frac{1}{2}m\omega^2 a^2 = \frac{1}{2} \times 0.5 \times 40 \times (0.2)^2 = 0.4\ \text{J}.$$

(c) The maximum speed occurs at the centre: $v_{\max} = a\omega = 0.2 \times 2\sqrt{10} = 0.4\sqrt{10} \approx 1.26\ \text{m s}^{-1}$.

2.2 The Simple Pendulum

The simple pendulum is the historical icon of SHM. It consists of a particle (the bob) of mass m attached to a light inextensible string of length l, swinging in a vertical plane about a fixed point.



Simple pendulum displaced through an angle θ. The tension T has no tangential component; the restoring force along the arc is $-mg\sin\theta$.

Let θ be the angular displacement from the vertical, measured in **radians**. The bob moves on a circular arc of radius l; its tangential acceleration is $l\ddot{\theta}$. The tension is perpendicular to the motion, so the only tangential force is the component of the weight, $-mg\sin\theta$ (negative because

it acts towards $\theta = 0$). Newton's second law along the tangent gives

$$ml\ddot{\theta} = -mg\sin\theta \quad \Rightarrow \quad \ddot{\theta} = -\frac{g}{l}\sin\theta.$$

This is not exactly SHM, because $\sin\theta \neq \theta$ in general. However, for small oscillations (in practice $\theta \lesssim 10^\circ \approx 0.17$ rad) we may use $\sin\theta \approx \theta$, giving

$$\ddot{\theta} \approx -\frac{g}{l}\theta,$$

which is the SHM equation with $\omega^2 = \dfrac{g}{l}$.

PROPERTY **Period of a Simple Pendulum**

For small oscillations, a simple pendulum of length l performs approximate SHM with period

$$T = 2\pi\sqrt{\frac{l}{g}}.$$

The period is independent of the mass m and of the (small) amplitude: the oscillations are isochronous.

WARNING **Radians Only!**

The approximation $\sin\theta \approx \theta$ is only valid when θ is measured in radians. In degrees the derivative of $\sin\theta$ is not $\cos\theta$, and the whole derivation collapses. Always convert angles to radians before applying any small-angle approximation.

DID YOU KNOW? **The Seconds Pendulum**

A seconds pendulum is one whose half-period (the time of one swing) is exactly one second, i.e. $T = 2$ s. Its length is $l = g/\pi^2$, about 0.99 m in Cameroon. Pendulum clocks based on this principle were the most accurate timekeepers in the world for nearly three centuries, and pendulum measurements were historically used to map the variation of g over the Earth's surface.

WORKED EXAMPLE 4 **Determining g with a pendulum**

In an experiment to determine g, a student measures the time for 20 complete oscillations of a simple pendulum of length 0.800 m as 36.2 s. Calculate the value of g obtained, and suggest two reasons why it might differ from the accepted value.

SOLUTION

The period is $T = \dfrac{36.2}{20} = 1.81\,\text{s}$. From $T = 2\pi\sqrt{l/g}$,

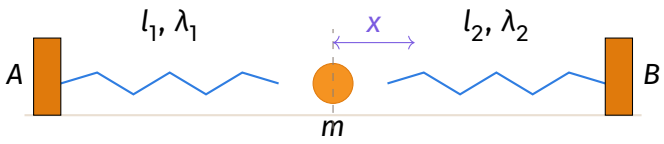
$$g = \frac{4\pi^2 l}{T^2} = \frac{4\pi^2 \times 0.800}{(1.81)^2} \approx 9.64\,\text{m s}^{-2}.$$

Possible reasons for the discrepancy:

1. The amplitude may have been too large; the true period increases slightly with amplitude, so the measured T is too large and g comes out too small.
2. Timing errors: reaction time in starting and stopping the stopwatch (reduced by timing many oscillations, but not eliminated).
3. The string has mass and the bob is not a point mass, so the system behaves as a compound pendulum with a slightly different period.

2.3 Oscillations with Two Springs or Elastic Strings

Many GCE problems involve a particle attached to two springs or elastic strings on a smooth horizontal surface. The key step is always to find the **net restoring force** as a function of the displacement x from the equilibrium position.



Particle of mass m between two light elastic strings fixed at A and B on a smooth table, displaced a distance x to the right of equilibrium.

Particle between two fixed points (the standard GCE configuration). Let the left string have natural length l_1 and modulus λ_1, the right string l_2 and λ_2, and let $AB = L$. At equilibrium the extensions e_1, e_2 satisfy

$$(l_1 + e_1) + (l_2 + e_2) = L, \qquad \frac{\lambda_1 e_1}{l_1} = \frac{\lambda_2 e_2}{l_2} \quad \text{(tensions balance)}.$$

Displace the particle a distance x to the right. The new extensions are $e_1 + x$ (left) and $e_2 - x$ (right). The net force, taking right as positive, is

$$F = \frac{\lambda_2(e_2 - x)}{l_2} - \frac{\lambda_1(e_1 + x)}{l_1} = \underbrace{\left(\frac{\lambda_2 e_2}{l_2} - \frac{\lambda_1 e_1}{l_1}\right)}_{=\,0\ \text{(equilibrium)}} - \left(\frac{\lambda_1}{l_1} + \frac{\lambda_2}{l_2}\right)x.$$

The constant terms cancel — this always happens when displacement is measured from equi-

librium. The motion is therefore SHM with

$$\omega^2 = \frac{\lambda_1}{ml_1} + \frac{\lambda_2}{ml_2}.$$

For identical strings ($\lambda_1 = \lambda_2 = \lambda$, $l_1 = l_2 = l$), $\omega^2 = \frac{2\lambda}{ml}$.

WARNING **Strings can go slack**

An elastic string (unlike a spring) cannot push: it exerts a force only when taut. The formula above is valid only while both strings remain taut, i.e. for $|x| \le \min(e_1, e_2)$. If the amplitude exceeds the smaller equilibrium extension, one string goes slack during the motion and the problem becomes an "interrupted motion" problem (see below). Always check this condition.

METHOD **Analysing Two-String Systems**

1. Draw a clear diagram showing the equilibrium position and a displaced position x.
2. Find the equilibrium extensions e_1, e_2 using the geometry and the balance of tensions.
3. Write the tensions in the displaced position in terms of x.
4. Write the net force $F(x)$; the constant (equilibrium) terms must cancel.
5. Read off ω^2 as the coefficient of $-x/m$, then $T = 2\pi/\omega$.
6. Check that both strings remain taut throughout the motion.

WORKED EXAMPLE 5 **Two strings on a smooth table**

A particle of mass 0.2 kg is attached to two light elastic strings of natural lengths 1 m and 1.5 m and moduli 10 N and 20 N respectively. The other ends are fixed to points A and B on a smooth horizontal table, where $AB = 4$ m. The particle is displaced 0.3 m from its equilibrium position and released from rest. Show that the motion is SHM and find its period.

SOLUTION

Let e_1, e_2 be the equilibrium extensions. Geometry: $(1 + e_1) + (1.5 + e_2) = 4 \Rightarrow e_1 + e_2 = 1.5$. Tension balance: $\frac{10e_1}{1} = \frac{20e_2}{1.5} \Rightarrow 10e_1 = \frac{40}{3}e_2 \Rightarrow e_1 = \frac{4}{3}e_2$. Substituting: $\frac{4}{3}e_2 + e_2 = 1.5 \Rightarrow \frac{7}{3}e_2 = 1.5 \Rightarrow e_2 = \frac{9}{14}$ m, $e_1 = \frac{6}{7}$ m. Both strings remain taut since the amplitude

$0.3 < \min(e_1, e_2) = 9/14 \approx 0.64$ m. Displace x to the right: extensions are $e_1 + x$ and $e_2 - x$, so

$$T_1 = 10(e_1 + x), \qquad T_2 = \frac{20}{1.5}(e_2 - x) = \frac{40}{3}(e_2 - x).$$

Net force (taking right as positive):

$$F = T_2 - T_1 = \frac{40}{3}e_2 - 10e_1 - \left(10 + \frac{40}{3}\right)x = -\frac{70}{3}x,$$

since $\frac{40}{3}e_2 = 10e_1$ at equilibrium. Hence

$$0.2\,\ddot{x} = -\frac{70}{3}x \quad \Rightarrow \quad \ddot{x} = -\frac{350}{3}x,$$

which is SHM with $\omega^2 = \dfrac{350}{3}$, so $\omega = \sqrt{350/3} \approx 10.8$ rad s^{-1} and

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{3}{350}} \approx 0.58\text{ s}.$$

2.4 Interrupted Motion: Strings Going Slack

A favourite GCE theme: the SHM is interrupted because a string goes slack, or the particle leaves a platform, or a collision occurs. The governing principle is:

KEY POINTS **Interrupted Motion Principle**

While the string is taut (or the spring is attached), the motion is SHM about the equilibrium position. When a string goes slack, its tension vanishes, the restoring force disappears, and the particle moves with constant velocity (horizontal motion) or as a projectile / under gravity alone (vertical motion) until the string becomes taut again. The two phases must be analysed separately.

WARNING **Slack string means no SHM**

The instant a string goes slack, $\ddot{x} = -\omega^2 x$ ceases to apply. Do not continue to use SHM formulas (such as $v = \omega\sqrt{a^2 - x^2}$) beyond the slack point. Use the velocity at the slack point as the initial velocity for the free phase.

WORKED EXAMPLE 6 **Horizontal string going slack**

A particle P of mass 0.5 kg is attached to one end of a light elastic string of natural length

1 m and modulus 20 N. The other end is fixed to a point O of a smooth horizontal table. The particle is held with the string stretched to a length 1.5 m and released from rest. Find (a) the time until the string first becomes slack, (b) the speed of P at that instant, (c) the distance travelled by P in the 0.2 s after the string goes slack.

SOLUTION

Measure x as the extension of the string. While taut, $\ddot{x} = -\dfrac{\lambda}{ml}x$ with

$$\omega^2 = \frac{\lambda}{ml} = \frac{20}{0.5 \times 1} = 40 \quad \Rightarrow \quad \omega = 2\sqrt{10}\ \text{rad s}^{-1}.$$

The amplitude is $a = 0.5$ m and, since P is released from rest at maximum extension, $x = 0.5\cos(\omega t)$. (a) The string goes slack when the extension first becomes zero: $\cos(\omega t) = 0 \Rightarrow \omega t = \dfrac{\pi}{2}$, so

$$t = \frac{\pi}{2\omega} = \frac{\pi}{4\sqrt{10}} \approx 0.248\ \text{s}.$$

(b) $\dot{x} = -0.5\,\omega\sin(\omega t)$; at $\omega t = \pi/2$, $|\dot{x}| = 0.5\omega = \sqrt{10} \approx 3.16\ \text{m s}^{-1}$. (c) Once slack, there is no horizontal force on P (smooth table), so it moves with constant speed $\sqrt{10}\ \text{m s}^{-1}$. In 0.2 s it travels

$$s = \sqrt{10} \times 0.2 \approx 0.63\ \text{m}.$$

WORKED EXAMPLE 7 **Vertical string going slack**

A particle of mass 0.2 kg hangs in equilibrium from a light elastic string of natural length 0.5 m and modulus 10 N. It is pulled down a further 0.3 m and released from rest. Find the time from release until the string first becomes slack. (Take $g = 9.8\ \text{m s}^{-2}$.)

SOLUTION

The equilibrium extension is $e = \dfrac{mgl}{\lambda} = \dfrac{0.2 \times 9.8 \times 0.5}{10} = 0.098$ m. About equilibrium, $\omega^2 = \dfrac{\lambda}{ml} = \dfrac{10}{0.2 \times 0.5} = 100$, so $\omega = 10\ \text{rad s}^{-1}$. Take x downwards from equilibrium; then $x = 0.3\cos(10t)$ (released from rest at $x = +0.3$). The total extension of the string is $e + x$, so the tension is

$$T = \frac{\lambda}{l}(e + x) = 20(0.098 + x).$$

The string goes slack when $T = 0$, i.e. $x = -0.098$ m (0.098 m above equilibrium). Note this point is reached because the amplitude $0.3 > e = 0.098$. Solve $0.3\cos(10t) = -0.098$: $\cos(10t) = -0.3267$, so the first positive solution is

$$10t = \arccos(-0.3267) \approx 1.904\ \text{rad} \quad \Rightarrow \quad t \approx 0.190\ \text{s}.$$

WORKED EXAMPLE 8 **One of two strings going slack**

A particle of mass 0.1 kg is attached to two identical light elastic strings, each of natural length 0.5 m and modulus 5 N. The other ends are fixed to points A and B on a smooth horizontal table with $AB = 2$ m. The particle is displaced 0.4 m from the centre towards B and released from rest. Show that one string goes slack during the motion, and find the speed of the particle at the instant of slackening.

SOLUTION

By symmetry the equilibrium position is the midpoint, where each string has length 1 m and extension $e = 0.5$ m. While both strings are taut, displacement x from the centre gives (identical strings)

$$\omega^2 = \frac{2\lambda}{ml} = \frac{2 \times 5}{0.1 \times 0.5} = 200 \quad \Rightarrow \quad \omega = 10\sqrt{2}\ \text{rad s}^{-1}.$$

The amplitude is $a = 0.4$ m. A string goes slack when its extension vanishes, i.e. when $|x| = e = 0.5$ m. Since $a = 0.4 < 0.5$, neither string goes slack: the motion is pure SHM throughout, and the premise fails.

Corrected data: suppose instead the particle is displaced 0.6 m towards B. Then $a = 0.6 > e = 0.5$, so the string attached to A goes slack when $x = +0.5$ m (moving towards B). Using $v^2 = \omega^2(a^2 - x^2)$,

$$v^2 = 200\,(0.6^2 - 0.5^2) = 200 \times 0.11 = 22 \quad \Rightarrow \quad v = \sqrt{22} \approx 4.69\ \text{m s}^{-1}.$$

Thereafter the particle continues under the tension of the B-string alone, which is a new SHM about a new centre — a typical extension for the final part of such a question.

2.5 Time Taken to Travel Between Two Points

A direct and very common examination task: given two displacements x_1 and x_2 measured from the centre of oscillation, find the time taken to travel from one to the other.

METHOD **Time from x_1 to x_2 in SHM**

1. Choose the form of the solution to match the initial condition: $x = a\cos(\omega t)$ if the particle starts at an extreme ($x = \pm a$ at $t = 0$); $x = a\sin(\omega t)$ if it starts at the centre moving positively; otherwise $x = a\cos(\omega t + \varepsilon)$.
2. Find the phase angles θ_1, θ_2 corresponding to x_1 and x_2, choosing the quadrants consistent with the direction of motion (the phase increases steadily with time).
3. The time taken is $\Delta t = \dfrac{|\theta_2 - \theta_1|}{\omega}$.

WORKED EXAMPLE 9 **Time between two points**

A particle performs SHM with amplitude 0.4 m and period 2 s. It is released from rest at the positive extreme. Find the time taken to travel (a) from $x = 0.4$ to $x = 0.2$, (b) from $x = 0.2$ to $x = -0.2$, (c) from $x = 0.2$ to the centre.

SOLUTION

$\omega = \dfrac{2\pi}{T} = \pi$ rad s^{-1} and $x = 0.4\cos(\pi t)$, with the particle moving towards the centre for $0 < t < 1$. (a) $x_1 = 0.4 \Rightarrow \pi t_1 = 0$; $x_2 = 0.2 \Rightarrow \cos(\pi t_2) = 0.5 \Rightarrow \pi t_2 = \dfrac{\pi}{3}$. Time $= \dfrac{\pi/3 - 0}{\pi} = \dfrac{1}{3}$ s. (b) $\pi t_1 = \dfrac{\pi}{3}$; $x_2 = -0.2 \Rightarrow \cos(\pi t_2) = -0.5 \Rightarrow \pi t_2 = \dfrac{2\pi}{3}$ (still in the first half-oscillation, moving negatively). Time $= \dfrac{2\pi/3 - \pi/3}{\pi} = \dfrac{1}{3}$ s. (c) $\pi t_1 = \dfrac{\pi}{3}$; $x_2 = 0 \Rightarrow \pi t_2 = \dfrac{\pi}{2}$. Time $= \dfrac{\pi/2 - \pi/3}{\pi} = \dfrac{1}{6}$ s.

Note the check: (b) = (a) + 2×(c)? Indeed $\frac{1}{3} = \frac{1}{3}$? No — the correct decomposition is (b) = (c) + time from centre to −0.2, and by symmetry of the cosine the time from 0.2 to 0 equals the time from 0 to −0.2, so (b) = 2×(c) = $\frac{1}{3}$ s. Consistent.

2.6 Approximate Simple Harmonic Motion

Many systems are not exactly simple harmonic but become so for small displacements. The recipe is always the same:

METHOD **Proving Approximate SHM**

1. Identify the equilibrium position.
2. Displace the system by a small amount x (or angle θ).
3. Write the exact equation of motion (Newton's second law, or torque = $I\ddot{\theta}$).
4. Linearise using the small-quantity approximations: $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1$ (to first order), $(1 + x)^n \approx 1 + nx$.
5. Show the equation reduces to $\ddot{x} = -\omega^2 x$ (or $\ddot{\theta} = -\omega^2\theta$) and identify ω^2.

WORKED EXAMPLE 10 **Bead on a smooth circular wire**

A bead of mass m is threaded on a smooth circular wire of radius R fixed in a vertical plane. The bead is slightly displaced from the lowest point. Show that it performs approximate SHM and find the period.

SOLUTION

Let θ be the angular displacement from the downward vertical. The tangential component of the weight is $-mg\sin\theta$ and the tangential acceleration is $R\ddot{\theta}$, so

$$mR\ddot{\theta} = -mg\sin\theta \quad \Rightarrow \quad \ddot{\theta} = -\frac{g}{R}\sin\theta.$$

For small θ (radians), $\sin\theta \approx \theta$: $\ddot{\theta} \approx -\frac{g}{R}\theta$ — SHM with $\omega^2 = \frac{g}{R}$, hence

$$T = 2\pi\sqrt{\frac{R}{g}}.$$

This is identical to a simple pendulum of length R: the wire supplies the constraint that the string supplied before.

WORKED EXAMPLE 11 **Floating cylinder**

A uniform solid cylinder of cross-sectional area A, height h and density ρ floats vertically in a liquid of density σ ($\rho < \sigma$). It is pushed down a small distance x and released. Show that the motion is SHM and find the period.

SOLUTION

At equilibrium, weight = upthrust (Archimedes): $\rho Ahg = \sigma Ah_0 g$, where h_0 is the submerged depth, so $h_0 = \frac{\rho}{\sigma}h$. Push the cylinder down by x (small enough that it remains partially immersed and not fully submerged). The submerged depth becomes $h_0 + x$, so the upthrust is $\sigma A(h_0 + x)g$. Taking downwards as positive, the net force is

$$F = \underbrace{\rho Ahg - \sigma Ah_0 g}_{=0\ \text{(equilibrium)}} - \sigma Ag\,x = -\sigma Ag\,x.$$

With mass $m = \rho Ah$,

$$\rho Ah\,\ddot{x} = -\sigma Ag\,x \quad \Rightarrow \quad \ddot{x} = -\frac{\sigma g}{\rho h}x,$$

SHM with $\omega^2 = \frac{\sigma g}{\rho h}$, so $T = 2\pi\sqrt{\frac{\rho h}{\sigma g}}$.

WARNING **Approximate SHM Checklist**

When asked to "show that the motion is approximately simple harmonic":

- State explicitly the approximation used (e.g. $\sin\theta \approx \theta$, θ in radians).

- Show that the restoring force (or torque) is proportional to the displacement and directed towards equilibrium.
- Identify ω^2 clearly and state the period if required.
- Mention that the result requires the amplitude to be small.

KEY POINTS **Key Formulas for this Section**

- Energy: $E = \frac{1}{2}m\omega^2 a^2$, $KE = \frac{1}{2}m\omega^2(a^2 - x^2)$, $PE = \frac{1}{2}m\omega^2 x^2$.
- Simple pendulum (small oscillations): $T = 2\pi\sqrt{\frac{l}{g}}$.
- Particle between two strings/springs: $\omega^2 = \frac{\lambda_1}{ml_1} + \frac{\lambda_2}{ml_2}$ (both taut).
- Time from x_1 to x_2: match phase angles in $x = a\cos(\omega t)$ or $a\sin(\omega t)$; $\Delta t = \Delta\theta/\omega$.
- Slack string $\Rightarrow$ constant velocity (horizontal) or motion under gravity alone (vertical).
- Approximate SHM: linearise with $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $(1+x)^n \approx 1 + nx$.

EXERCISE 4 **GCE-Style Practice**

1. A particle of mass 0.3 kg is attached to a light elastic string of natural length 0.8 m and modulus 12 N on a smooth horizontal table. The particle is pulled until the extension is 0.4 m and released from rest. (a) Show that the motion is SHM and find ω. (b) Find the total energy. (c) Find the speed when the extension is 0.1 m. (d) Find the time taken to go from the point of release to the point where the extension is 0.1 m.

2. A simple pendulum of length 1.2 m makes small oscillations where $g = 9.8\ \text{m s}^{-2}$. (a) Find its period. (b) The pendulum is taken to a planet where $g = 4.0\ \text{m s}^{-2}$. By what factor does the period change? (c) If the amplitude is 5°, is the small-angle approximation justified? (Use $\sin 5^\circ \approx 0.0872$ and $5^\circ \approx 0.0873$ rad.)

3. A particle of mass m is attached to two identical light elastic strings, each of natural length l and modulus λ. The other ends are fixed to points A and B on a smooth horizontal table with $AB = 4l$. The particle is held at the midpoint of AB and released from rest. Show that the motion is SHM and find the period in terms of m, λ, l.

4. A uniform rod of length $2L$ and mass M is pivoted smoothly at one end and hangs vertically. It is displaced through a small angle and released. By considering the torque about the pivot, show that the rod performs approximate SHM and find its period.

(Moment of inertia of a uniform rod of length $2L$ about one end: $I = \frac{4}{3}ML^2$; the centre of mass is at distance L from the pivot.)

5. A cylindrical buoy of mass m and cross-sectional area A floats vertically in water of density ρ. It is pushed down a small distance and released. Show that it performs SHM and find the period. If a bird of mass m_b lands gently on the buoy, how does the period change?

SOLUTION **Exercise 1**

(a) With extension x, the only horizontal force is the tension $T = \frac{\lambda x}{l}$ directed towards the natural-length position, so $m\ddot{x} = -\frac{\lambda}{l}x$, i.e. $\ddot{x} = -\frac{\lambda}{ml}x$: SHM. $\omega^2 = \frac{12}{0.3 \times 0.8} = 50$, $\omega = 5\sqrt{2} \approx 7.07\,\text{rad s}^{-1}$. **(b)** $E = \frac{1}{2}m\omega^2 a^2 = \frac{1}{2} \times 0.3 \times 50 \times (0.4)^2 = 1.2\,\text{J}$. **(c)** $v^2 = \omega^2(a^2 - x^2) = 50(0.16 - 0.01) = 7.5 \Rightarrow v = \sqrt{7.5} \approx 2.74\,\text{m s}^{-1}$. **(d)** $x = 0.4\cos(\omega t)$. At $x = 0.1$: $\cos(\omega t) = 0.25 \Rightarrow \omega t = \arccos(0.25) \approx 1.318\,\text{rad}$, so $t = \frac{1.318}{5\sqrt{2}} \approx 0.186\,\text{s}$.

SOLUTION **Exercise 2**

(a) $T = 2\pi\sqrt{\frac{1.2}{9.8}} \approx 2.20\,\text{s}$. **(b)** $T \propto 1/\sqrt{g}$, so the period is multiplied by $\sqrt{9.8/4.0} \approx 1.57$ (it increases). **(c)** $\sin 5^\circ \approx 0.0872$ while $\theta = 0.0873\,\text{rad}$: a relative error of about 0.1%. The approximation is excellent.

SOLUTION **Exercise 3**

At the midpoint each string has length $2l$ and extension l; the tensions are equal, so the midpoint is the equilibrium position. Displace x towards B: extensions are $l + x$ and $l - x$ (both positive for $|x| < l$, so both strings stay taut).

$$T_L = \frac{\lambda}{l}(l + x), \qquad T_R = \frac{\lambda}{l}(l - x).$$

Net force towards the centre: $F = T_R - T_L = -\frac{2\lambda}{l}x$. Hence $\ddot{x} = -\frac{2\lambda}{ml}x$: SHM with $\omega^2 = \frac{2\lambda}{ml}$ and

$$T = 2\pi\sqrt{\frac{ml}{2\lambda}}.$$

SOLUTION **Exercise 4**

At angular displacement θ, the weight Mg acts at the centre of mass, distance L from the pivot, giving a restoring torque $\tau = -MgL\sin\theta \approx -MgL\theta$ for small θ. With $I = \frac{4}{3}ML^2$,

$$\frac{4}{3}ML^2\,\ddot{\theta} = -MgL\,\theta \quad \Rightarrow \quad \ddot{\theta} = -\frac{3g}{4L}\theta.$$

SHM with $\omega^2 = \dfrac{3g}{4L}$, so $T = 2\pi\sqrt{\dfrac{4L}{3g}}$.

SOLUTION **Exercise 5**

At equilibrium $mg = \rho A h_0 g$, giving submerged depth $h_0 = \dfrac{m}{\rho A}$. Pushed down by x, the extra upthrust is $\rho A g x$ upwards, so $m\ddot{x} = -\rho A g\, x$: SHM with $\omega^2 = \dfrac{\rho A g}{m}$ and

$$T = 2\pi\sqrt{\frac{m}{\rho A g}}.$$

With the bird, the mass becomes $m + m_b$ while ρ, A, g are unchanged, so

$$T' = 2\pi\sqrt{\frac{m + m_b}{\rho A g}} = T\sqrt{\frac{m + m_b}{m}} > T\ :$$

the period increases.

3 DAMPED HARMONIC MOTION

In the previous section we studied simple harmonic motion in its ideal form: a mass on a spring, a pendulum swinging in a vacuum, oscillations that continue forever with constant amplitude because no energy is ever lost. But walk through any mechanic's workshop in Douala or Bamenda and push a car down on its suspension: the car bounces a few times and then settles. A pendulum clock eventually stops unless it is wound. A plucked guitar string fades into silence. Real oscillators lose energy, and their motion dies away. This section builds the mathematics of that reality: **damped harmonic motion**. It is one of the most rewarding parts of the syllabus, because the same differential equation describes car suspensions, earthquake-resistant buildings and the needle of an analogue voltmeter.

3.1 The Physical Origin of Damping

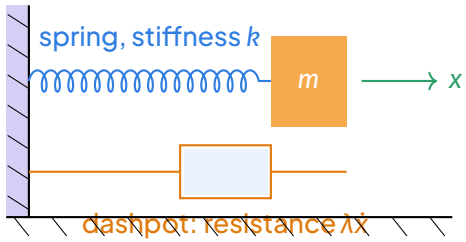
When a body oscillates in a real medium, it experiences a **resistive force** opposing its motion. The sources are many:

- **Air resistance:** a swinging pendulum pushes air aside and is slowed by it.
- **Fluid friction:** a **dashpot** (a piston moving in a cylinder of oil, exactly the device inside a car's **shock absorber**) converts mechanical energy into heat.
- **Internal friction:** the repeated flexing of a spring warms it slightly.

For the moderate speeds met in oscillating systems, experiments show that the resistance is well modelled as proportional to the velocity and directed opposite to it:

$$R = -\lambda \dot{x}, \qquad \lambda > 0,$$

where λ is the **damping coefficient** (in kg s^{-1}). The minus sign is essential: the resistance always acts against the direction of motion, whatever that direction is.



A mass m attached to a spring of stiffness k and a dashpot providing resistance $\lambda \dot{x}$.

3.2 Setting Up the Equation of Motion

Consider a particle of mass m attached to a spring of natural length l and modulus of elasticity λ_e (stiffness $k = \lambda_e / l$), moving horizontally on a smooth table, subject to a resistance $\lambda \dot{x}$. Measuring x from the equilibrium position (so that the weight and any static extension cancel, exactly as in our SHM work), Newton's second law gives

$$m\ddot{x} = -kx - \lambda \dot{x}.$$

Dividing by m and renaming the constants, we obtain the standard form.

DEFINITION **Damped harmonic motion**

A particle performs **damped harmonic motion** when its displacement x from equilibrium satisfies

$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0,$$

where $\omega^2 = \dfrac{k_{\text{spring}}}{m}$ is the square of the **natural angular frequency** (the frequency the system would have without damping) and $k = \dfrac{\lambda}{2m} > 0$ measures the **strength of the damping**. When $k = 0$ the equation reduces to simple harmonic motion, $\ddot{x} + \omega^2 x = 0$.

WARNING **Notation alert**

In the standard form $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$, the letter k denotes the damping parameter $\lambda/(2m)$, not the spring stiffness. Some textbooks write $2n$ or 2γ instead of $2k$. Always check the notation of the question before quoting formulae, and state your own notation clearly at the start of your solution.

3.3 Solving the Equation: the Auxiliary Equation

The equation is linear with constant coefficients, so we try $x = e^{rt}$. Substituting and dividing by $e^{rt} \neq 0$ gives the **auxiliary equation**

$$r^2 + 2kr + \omega^2 = 0, \qquad \text{with roots} \qquad r = -k \pm \sqrt{k^2 - \omega^2}.$$

Everything now hinges on the sign of the discriminant $k^2 - \omega^2$, i.e. on how the damping k compares with the natural frequency ω.

PROPERTY **The three regimes of damping — proof examinable**

The nature of the motion governed by $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ depends on the sign of $k^2 - \omega^2$:

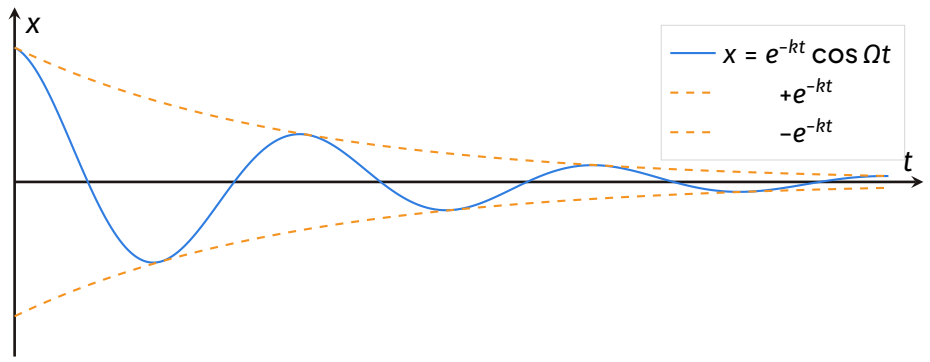
- $k < \omega$ (**light** or under damping): complex roots $-k \pm i\Omega$; the motion oscillates with decaying amplitude.
- $k = \omega$ (**critical** damping): repeated root $-k$; the motion returns to equilibrium as fast as possible without oscillating.
- $k > \omega$ (**heavy** or over damping): two real negative roots; the motion returns to equilibrium slowly, without oscillating.

3.4 Light Damping ($k < \omega$): Oscillations that Fade Away

When $k < \omega$, write $\Omega = \sqrt{\omega^2 - k^2} > 0$. The roots are $r = -k \pm i\Omega$, and the general solution is

$$\boxed{x = e^{-kt}\left(A\cos \Omega t + B \sin \Omega t\right), \qquad \Omega = \sqrt{\omega^2 - k^2}.}$$

This is a beautiful formula: it is SHM of angular frequency Ω multiplied by a dying exponential. The motion still crosses equilibrium regularly, but each swing is smaller than the last. The curve is trapped between the two **exponential envelopes** $x = \pm a\, e^{-kt}$.



Lightly damped oscillation: the curve oscillates inside the dashed exponential envelopes $\pm ae^{-kt}$.

The damped motion is not strictly periodic (the amplitude changes), but successive passages through equilibrium in the same direction are separated by a constant time, the **damped period**:

$$T_d = \frac{2\pi}{\Omega} = \frac{2\pi}{\sqrt{\omega^2 - k^2}}.$$

WARNING **The damped period is longer**

The period of a lightly damped oscillator is $\frac{2\pi}{\sqrt{\omega^2 - k^2}}$, not $\frac{2\pi}{\omega}$. Since $\sqrt{\omega^2 - k^2} < \omega$, damping always lengthens the period. Quoting $2\pi/\omega$ for a damped system is a classic GCE error.

EXEMPLE **Finding the constants for light damping**

A particle moves with $\ddot{x} + 2\dot{x} + 10x = 0$, with $x = 2$ and $\dot{x} = 0$ at $t = 0$. Here $k = 1$, $\omega^2 = 10$, so $k^2 - \omega^2 = -9 < 0$: light damping with $\Omega = \sqrt{10 - 1} = 3$. Hence

$$x = e^{-t}(A\cos 3t + B\sin 3t).$$

From $x(0) = 2$: $A = 2$. Differentiating,

$$\dot{x} = e^{-t}\left[(-3A\sin 3t + 3B\cos 3t) - (A\cos 3t + B\sin 3t)\right],$$

so $\dot{x}(0) = 3B - A = 0$, giving $B = \frac{2}{3}$. Therefore

$$x = e^{-t}\left(2\cos 3t + \tfrac{2}{3}\sin 3t\right), \qquad T_d = \frac{2\pi}{3}\text{ s}.$$

3.5 Critical Damping ($k = \omega$): the Fastest Non-Oscillatory Return

When $k = \omega$ the auxiliary equation has the repeated root $r = -k$. A repeated root of a linear equa-

tion always contributes solutions e^{-kt} and $t\,e^{-kt}$, so

$$\boxed{x = (A + Bt)e^{-kt}.}$$

The factor e^{-kt} eventually dominates the linear factor $A + Bt$, so $x \to 0$. The particle returns to equilibrium without oscillating (it may cross equilibrium at most once, when A and B have opposite signs), and it does so faster than any over-damped system. This is exactly how a good car shock absorber is tuned: after hitting a pothole on the Yaoundé–Bafoussam road, the wheel should settle immediately, not bounce.

WARNING **GCE exam tip — the repeated root**

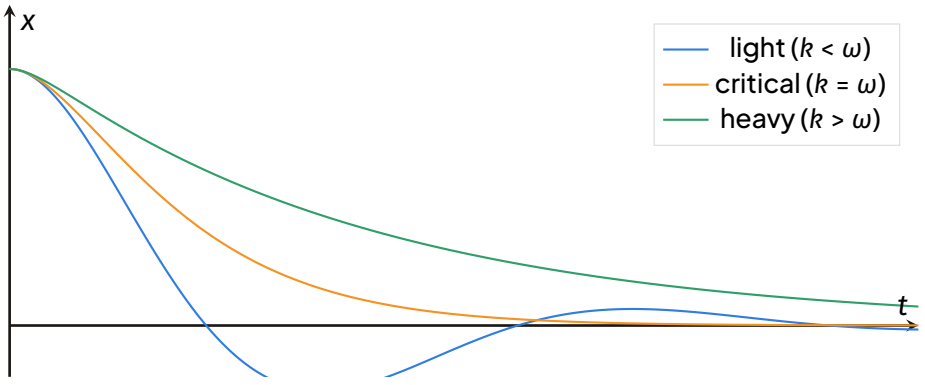
For critical damping the solution is $(A + Bt)e^{-kt}$. The extra factor t comes from the repeated root of the auxiliary equation. Writing only Ae^{-kt} (a single exponential) is the most common error in this topic and costs nearly all the marks for the solution, because the general solution then has only one arbitrary constant and cannot satisfy two initial conditions.

3.6 Heavy Damping ($k > \omega$): a Slow Creep Back

When $k > \omega$, the roots are real and distinct: $r = -k \pm \gamma$ with $\gamma = \sqrt{k^2 - \omega^2} < k$, so both roots are negative. The general solution is

$$\boxed{x = e^{-kt}\left(Ae^{\gamma t} + Be^{-\gamma t}\right) = Ae^{-(k-\gamma)t} + Be^{-(k+\gamma)t}, \qquad \gamma = \sqrt{k^2 - \omega^2}.}$$

Both exponentials decay; there is no oscillation at all. The slowest-decaying term, $e^{-(k-\gamma)t}$, governs the long-term behaviour, and since $k - \gamma$ is small when damping is heavy, the return to equilibrium is slower than in the critical case. A door-closer that is too stiff is over-damped: the door creeps shut frustratingly slowly.



The three regimes for the same initial conditions $x(0) = 1$, $\dot{x}(0) = 0$ (with $\omega = 1$). Critical damping returns to equilibrium fastest without oscillating.

3.7 The Logarithmic Decrement: Measuring Damping Experimentally

For a lightly damped oscillator, consider two successive maximum displacements on the same side, occurring one damped period T_d apart. Since the cosine factor repeats while the exponential decays,

$$\frac{x_n}{x_{n+1}} = \frac{e^{-kt}}{e^{-k(t+T_d)}} = e^{kT_d}, \qquad \text{so} \qquad \ln\left(\frac{x_n}{x_{n+1}}\right) = kT_d.$$

DEFINITION **Logarithmic decrement**

The **logarithmic decrement** of a lightly damped oscillation is

$$\Lambda = \ln\left(\frac{x_n}{x_{n+1}}\right) = kT_d = \frac{2\pi k}{\sqrt{\omega^2 - k^2}},$$

the natural logarithm of the ratio of two successive amplitudes on the same side. It is constant, so measuring any two successive amplitudes experimentally yields the damping constant k.

This is a GCE-useful extension: given a table of experimental amplitudes, compute $\ln(x_n / x_{n+1})$ for each pair, average, and deduce k. Note also the practical trick $\ln(x_0 / x_n) = n\Lambda$: using amplitudes many periods apart reduces measurement error.

3.8 Method and Worked Examination Problem

METHOD **Solving a damped harmonic motion problem**

1. Set up the equation from Newton's second law (resistance $-\lambda\dot{x}$, restoring force $-k_{\text{spring}}x$) and reduce it to the standard form $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$, identifying k and ω^2.
2. Write the auxiliary equation $r^2 + 2kr + \omega^2 = 0$ and compute $k^2 - \omega^2$ to identify the regime.
3. Write the standard solution for that regime:

$$\begin{cases} k < \omega: & x = e^{-kt}(A\cos\Omega t + B\sin\Omega t),\ \Omega = \sqrt{\omega^2 - k^2}, \\ k = \omega: & x = (A + Bt)e^{-kt}, \\ k > \omega: & x = Ae^{-(k-\gamma)t} + Be^{-(k+\gamma)t},\ \gamma = \sqrt{k^2 - \omega^2}. \end{cases}$$

4. Apply the initial conditions $x(0)$ and $\dot{x}(0)$ to find A and B (differentiate carefully using the product rule).
5. Answer the question asked: damped period, logarithmic decrement, number of crossings, time to halve the amplitude, etc.

WORKED EXAMPLE 12 **A complete GCE-style problem**

A particle P of mass 0.5 kg is attached to one end of a light elastic spring of stiffness 8 N m^{-1}, the other end being fixed. P moves in a straight line on a smooth horizontal table and experiences a resistance of magnitude $4v$ newtons, where v is its speed. The particle is pulled 0.3 m from equilibrium and released from rest.

(i) Show that the displacement x satisfies $\ddot{x} + 8\dot{x} + 16x = 0$.
(ii) Solve the equation and describe the motion.
(iii) Find the time at which the speed is greatest, and state whether P crosses the equilibrium position.

SOLUTION

(i) Newton's second law with restoring force $-8x$ and resistance $-4\dot{x}$:

$$0.5\,\ddot{x} = -8x - 4\dot{x} \implies \ddot{x} + 8\dot{x} + 16x = 0. \quad \blacksquare$$

(ii) Standard form: $2k = 8 \Rightarrow k = 4$; $\omega^2 = 16 \Rightarrow \omega = 4$. Since $k = \omega$, the damping is critical. Auxiliary equation: $r^2 + 8r + 16 = (r + 4)^2 = 0$, repeated root $r = -4$. General solution:

$$x = (A + Bt)e^{-4t}.$$

Initial conditions: $x(0) = 0.3 \Rightarrow A = 0.3$. Differentiate:

$$\dot{x} = Be^{-4t} - 4(A + Bt)e^{-4t}, \qquad \dot{x}(0) = B - 4A = 0 \Rightarrow B = 1.2.$$

$$\boxed{x = (0.3 + 1.2t)e^{-4t}}$$

The motion is non-oscillatory: P moves directly towards equilibrium, $x \to 0$ as $t \to \infty$.
(iii) Since $0.3 + 1.2t > 0$ for all $t \geq 0$, x never changes sign: P never crosses the equilibrium position. The speed is greatest when $\ddot{x} = 0$. From the equation, $\ddot{x} = -8\dot{x} - 16x$; setting $\ddot{x} = 0$:

$$\dot{x} = (1.2 - 4(0.3 + 1.2t))e^{-4t} = -4.8t\,e^{-4t},$$

so $\ddot{x} = -4.8e^{-4t} + 19.2t\,e^{-4t} = 0 \Rightarrow t = \dfrac{4.8}{19.2} = 0.25$ s. The greatest speed is $|\dot{x}(0.25)| = 4.8(0.25)e^{-1} = 1.2e^{-1} \approx 0.44$ m s^{-1}.

EXERCISE 5 **Light damping and the logarithmic decrement**

A particle performs damped harmonic motion with $\ddot{x} + 2\dot{x} + 26x = 0$. At $t = 0$, $x = 0.1$ m and $\dot{x} = 0$.

1. Identify the regime and find x in terms of t.
2. Find the damped period T_d.

3. Show that the ratio of successive maximum displacements is $e^{\pi/5}$, and find after how many complete oscillations the amplitude first falls below 0.01 m.

SOLUTION **Exercise 1**

1. $k = 1$, $\omega^2 = 26$; $k^2 - \omega^2 = -25 < 0$: light damping, $\Omega = \sqrt{26 - 1} = 5$. So $x = e^{-t}(A\cos 5t + B\sin 5t)$. $x(0) = 0.1 \Rightarrow A = 0.1$; $\dot{x}(0) = 5B - A = 0 \Rightarrow B = 0.02$. Hence $x = e^{-t}(0.1\cos 5t + 0.02\sin 5t)$.
2. $T_d = \dfrac{2\pi}{5}\,\text{s} \approx 1.26\,\text{s}$.
3. Ratio $= e^{kT_d} = e^{2\pi/5}$… note $kT_d = 1 \times \frac{2\pi}{5} = \frac{2\pi}{5}$, so the ratio is $e^{2\pi/5}$. Amplitude after n periods: $a_n = 0.1\,e^{-nT_d} = 0.1e^{-2\pi n/5}$. We need $e^{-2\pi n/5} < 0.1$, i.e. $\frac{2\pi n}{5} > \ln 10 \approx 2.303$, giving $n > \dfrac{5 \times 2.303}{2\pi} \approx 1.83$. So the amplitude first falls below 0.01 m after **2** complete oscillations.

KEY POINTS **Key points of the section**

- Damping adds a resistance $-\lambda\dot{x}$, giving $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ with $k = \lambda/(2m)$.
- The auxiliary equation $r^2 + 2kr + \omega^2 = 0$ decides everything via the sign of $k^2 - \omega^2$.
- Light damping ($k < \omega$): $x = e^{-kt}(A\cos\Omega t + B\sin\Omega t)$, $\Omega = \sqrt{\omega^2 - k^2}$, damped period $T_d = 2\pi/\Omega > 2\pi/\omega$.
- Critical damping ($k = \omega$): $x = (A + Bt)e^{-kt}$ — fastest return without oscillation; do not forget the t.
- Heavy damping ($k > \omega$): $x = Ae^{-(k-\gamma)t} + Be^{-(k+\gamma)t}$, $\gamma = \sqrt{k^2 - \omega^2}$ — slow, non-oscillatory.
- Logarithmic decrement: $\Lambda = \ln(x_n / x_{n+1}) = kT_d$; the practical tool for measuring damping.

DID YOU KNOW? **Damping all around us**

Engineers choose the damping regime deliberately. Car shock absorbers are designed to be very nearly critically damped so the wheels recover instantly from bumps. The needles of analogue ammeters and the moving parts of traditional balances are critically damped so the reading settles without swinging. Buildings in earthquake zones are fitted with giant tuned mass dampers — huge pendulums with dashpots — that absorb the oscillation energy of the structure. Conversely, a lightly damped system is wanted when oscillations must persist, as in the quartz crystal of a watch. The single equation of this section governs

them all.

We now have the complete picture: undamped SHM, the energy and pendulum results, and the three damping regimes. In the next and final section of this chapter we put all of this to work on applications and full GCE examination problems, including composite systems and questions that mix SHM with damping.

4 APPLICATIONS AND GCE PROBLEM-SOLVING

In the previous section we learnt that when a resistive force proportional to velocity acts on an oscillator, the equation of motion becomes $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$, and that the nature of the motion — underdamped, critically damped or overdamped — depends on the sign of $k^2 - \omega^2$. We now put the whole machinery of this chapter to work. Real systems — a car suspension, a market spring balance, a log bobbing on a river, a pendulum in the school laboratory — are all modelled by the equations we have derived. The GCE examiner's favourite skill is precisely this: translate a physical situation into a differential equation, solve it, and interpret the answer. This section trains exactly that skill.

4.1 The Modelling Checklist

Almost every structured question in this chapter follows the same logical path. Learn it as a routine.

METHOD **Modelling checklist for oscillation problems**

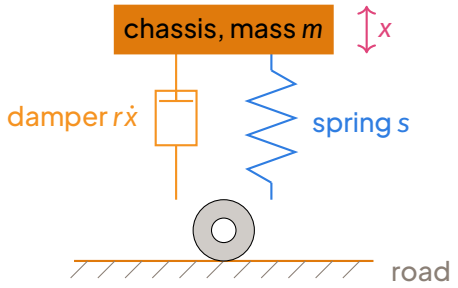
1. Find the equilibrium position first. Resolve forces with the system at rest. This gives a relation (e.g. $mg = \lambda e/l$ for a spring) that will simplify the equation later.
2. Define the displacement variable x from equilibrium, with a clearly stated positive direction.
3. List all forces in the displaced position: weight, tension or spring force (Hooke's law $T = \lambda(\text{extension})/l$), upthrust, resistance $-2mk\dot{x}$ if present.
4. Apply $F = ma$ along the positive direction: $m\ddot{x} = \sum F$.
5. Use the equilibrium relation to cancel the constant terms. What remains should be $\ddot{x} = -\omega^2 x$ (SHM) or $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ (damped).
6. Identify ω^2 (and k) by comparing coefficients.
7. Solve using the standard forms, then apply the initial conditions.
8. Interpret: period, amplitude, times, speeds — and state the limits of the model.

WARNING **The sign of the damping term**

A resistance proportional to velocity opposes the motion, so it must enter the equation with a **negative sign**: $-2mk\dot{x}$. If you obtain $\ddot{x} - 2k\dot{x} + \omega^2 x = 0$ (positive damping), the auxiliary equation gives roots with positive real part and the amplitude grows exponentially — physically absurd for a passive system. A positive sign is a reliable signal that you have made a sign error in your force diagram.

4.2 Vehicle Suspension: Why Near-Critical Damping?

A car wheel, together with the share of the chassis it carries, can be modelled as a mass m supported by a spring of stiffness s (force = $s\times$ compression) and a shock absorber (damper) exerting a resistance $r\dot{x}$.



Quarter-car model: the chassis mass m is connected to the wheel (and road) by a spring of stiffness s and a damper resisting with force $r\dot{x}$.

Measuring x from the equilibrium compression of the spring, the weight cancels the static spring force exactly as in the vertical mass–spring of Section 1, and the equation of motion is

$$m\ddot{x} = -sx - r\dot{x} \quad\Longrightarrow\quad \ddot{x} + 2k\dot{x} + \omega^2 x = 0, \qquad \omega^2 = \frac{s}{m}, \quad 2k = \frac{r}{m}.$$

When the wheel hits a pothole, the chassis is displaced and the suspension must bring it back to rest. The three regimes feel very different to the passenger:

- **Underdamped** ($k < \omega$): the car bounces several times after the bump — uncomfortable and dangerous, since the tyres lose grip during the oscillation.
- **Overdamped** ($k > \omega$): the return is slow and the suspension feels "dead"; the car cannot respond to the next bump in time.
- **Critically damped** ($k = \omega$): the chassis returns to equilibrium in the shortest possible time without oscillating.

This is why engineers tune shock absorbers to be **near-critical**: $k \approx \omega$, i.e. $r \approx 2\sqrt{ms}$. Critical damping is not a mathematical curiosity — it is a design target.

EXEMPLE **Tuning a shock absorber**

A car's suspension carries a mass $m = 300$ kg per wheel on a spring of stiffness $s = 27\,000\ \mathrm{N\,m^{-1}}$. Then

$$\omega = \sqrt{\frac{s}{m}} = \sqrt{90} \approx 9.49\ \mathrm{rad\,s^{-1}}.$$

Critical damping requires $k = \omega$, i.e. $r = 2m\omega \approx 2(300)(9.49) \approx 5\,690\ \mathrm{N\,s\,m^{-1}}$. A damper much weaker than this lets the car bounce; much stronger makes the ride harsh.

4.3 The Spring Scale and the Market Balance

A spring balance hanging in a Bamenda market stall is a vertical mass–spring system. When a bag of beans of unknown mass m is hung on the hook and set bobbing gently, the period of oscillation is

$$T = 2\pi\sqrt{\frac{m}{s}},$$

where s is the stiffness of the spring. Measuring T (time ten oscillations with a phone stopwatch and divide by ten) therefore measures the mass: $m = sT^2/(4\pi^2)$. This is the principle of the inertial balance, used where gravity is unreliable — for instance to "weigh" astronauts in orbit, where ordinary scales read zero but inertia is unchanged.

EXEMPLE **Mass from the period**

A spring scale oscillates with period $T = 0.80$ s when loaded with an unknown mass; its stiffness is $s = 50\ \mathrm{N\,m^{-1}}$. Then

$$m = \frac{sT^2}{4\pi^2} = \frac{50 \times 0.64}{4\pi^2} \approx 0.81\ \mathrm{kg}.$$

4.4 Buoyancy Oscillations: a Floating Cylinder

A cylindrical log of mass m and uniform cross-sectional area A floats vertically in water of density ρ. Push it down a little and release: it bobs up and down. Is this SHM? Let us prove it.

At equilibrium, the weight is balanced by the upthrust (Archimedes' principle): if d is the submerged depth at equilibrium, $mg = \rho g A d$. Now displace the cylinder a further distance x downwards from equilibrium. The submerged depth becomes $d + x$, so the upthrust is $\rho g A(d + x)$, and the resultant force (taking downwards as positive) is

$$m\ddot{x} = mg - \rho g A(d + x) = \underbrace{mg - \rho g A d}_{=\,0\ \text{(equilibrium)}} - \rho g A x = -\rho g A x.$$

PROPERTY **Vertical oscillations of a floating body**

A body of mass m and uniform cross-sectional area A, floating vertically in a liquid of den-

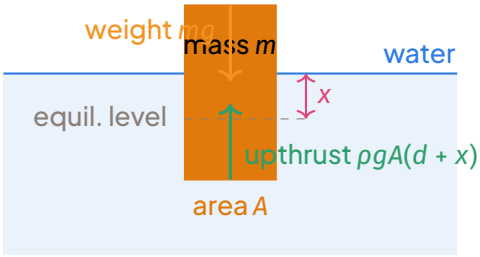
sity ρ and displaced by x from its floating equilibrium, satisfies

$$\ddot{x} = -\frac{\rho g A}{m}\,x,$$

and hence performs **simple harmonic motion** with

$$\omega^2 = \frac{\rho g A}{m}, \qquad T = 2\pi\sqrt{\frac{m}{\rho g A}}.$$

(The proof above — equilibrium condition plus Archimedes' principle — is examinable.)



A floating cylinder pushed down by x: the upthrust exceeds the equilibrium value by $\rho g A x$, giving a restoring force proportional to x.

WARNING **Units, units, units**

Always check units before substituting: ω in rad s^{-1}, modulus of elasticity λ in N, natural length and extension in m, density in kg m^{-3}, area in m^2. Mixing centimetres and metres is the single most frequent source of wrong periods at the GCE. Convert everything to SI units before writing the equation of motion.

4.5 The Piston and Crankshaft (Extension)

A car engine's piston is driven by a crank of radius r rotating at angular speed ω, connected to the piston by a rod of length L. Taking the crank angle as $\theta = \omega t$, geometry gives the piston displacement from the top of its stroke as

$$x = r(1 - \cos\omega t) + L - \sqrt{L^2 - r^2\sin^2\omega t}.$$

This is not exactly SHM. But in a real engine $r \ll L$, so $\sqrt{L^2 - r^2\sin^2\omega t} \approx L - \frac{r^2}{2L}\sin^2\omega t \approx L$, and

$$x \approx r(1 - \cos\omega t),$$

which is SHM of amplitude r about the mid-stroke position. Engineers use this approximation to estimate piston speeds and the forces on the connecting rod; the small correction term is

responsible for the high-frequency "buzz" of a fast engine. This result is an extension: you will not be asked to derive it, but it shows SHM appearing as an approximation inside real machines.

4.6 Measuring g with a Simple Pendulum

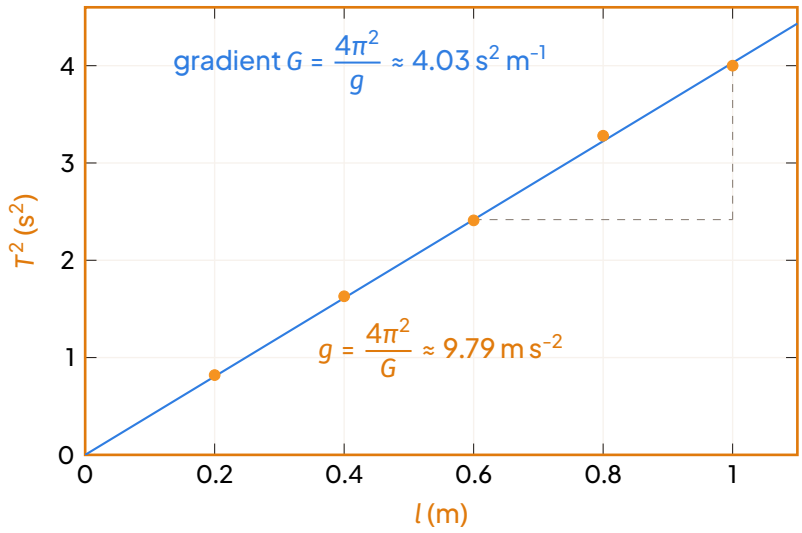
For small oscillations, a simple pendulum of length l has period

$$T = 2\pi\sqrt{\frac{l}{g}} \quad \Longrightarrow \quad T^2 = \frac{4\pi^2}{g}\,l.$$

This is a linear relation between T^2 and l, with gradient $4\pi^2/g$. This **linearisation** is the standard experimental strategy: instead of computing g from one measurement (one bad reading ruins everything), plot T^2 against l for several lengths and take the gradient of the best-fit line.

METHOD **Experimental determination of g**

1. Suspend a small dense bob on a light inextensible string; measure l from the support to the centre of the bob.
2. Displace the bob through a small angle (less than about 10°) and release.
3. Time 20 complete oscillations, repeat, and average; divide by 20 to get T.
4. Repeat for at least five different lengths.
5. Plot T^2 (vertical) against l (horizontal); draw the best-fit straight line through the origin.
6. Measure the gradient G; then $g = \dfrac{4\pi^2}{G}$.



Plot of T^2 against l for pendulum data. The gradient of the best-fit line is $4\pi^2/g$, so $g = 4\pi^2/G$.

WORKED EXAMPLE 13 **Computing g from the graph**

A student measures the gradient of her T^2–l graph as $G = 4.06\,\mathrm{s^2\,m^{-1}}$. Find the value of g she should quote.

SOLUTION

Since $T^2 = \frac{4\pi^2}{g} l$, the gradient is $G = \frac{4\pi^2}{g}$, hence

$$g = \frac{4\pi^2}{G} = \frac{4\pi^2}{4.06} = \frac{39.478}{4.06} \approx 9.72\,\mathrm{m\,s^{-2}}.$$

She should quote $g \approx 9.7\,\mathrm{m\,s^{-2}}$, and could comment that the small discrepancy from the accepted value may come from reaction time in the timing or from measuring l to the top rather than the centre of the bob.

4.7 SHM or Damped? Reading the Problem Statement

Before solving, decide which model the question intends. The wording tells you.

Clue in the statement	Mo	Equatio
“light resistance neglected”, “smooth”, no mention of friction	SHM	$\ddot{x} = -\omega^2 x$
“resistance proportional to speed”, “dashpot”, “shock absorber”, “damping force $2mkv$”	Damped	$\ddot{x} + 2k\dot{x} + \omega^2 x = 0$
“comes to rest without oscillating as quickly as possible”	Critical damping	$k = \omega$
“oscillations die away”, “amplitude decreases”	Under-damp	$k < \omega$

If a resistance is mentioned, it must appear in your force equation; if none is mentioned, do not invent one. Conversely, if a question later says “in practice the amplitude decreases, criticise the model”, the expected answer is that real resistance (air drag, internal friction) was neglected.

4.8 A Full GCE-Type Structured Problem

WORKED EXAMPLE 14 **Floating buoy (structured question)**

A cylindrical buoy of mass $m = 12$ kg and cross-sectional area $A = 0.030\ \text{m}^2$ floats vertically in sea water of density $\rho = 1025\ \text{kg m}^{-3}$. Take $g = 9.81\ \text{m s}^{-2}$.
(a) Show that, when the buoy is displaced vertically by x metres from equilibrium and resistance is neglected, $\ddot{x} = -25.1\,x$ approximately, and state the value of ω.
(b) The buoy is pushed down 0.20 m and released from rest. Find the period and the maximum speed.
(c) Find the time at which the buoy first passes through its equilibrium position.
(d) In practice the oscillations die away. Criticise the model.

SOLUTION

(a) At equilibrium, $mg = \rho g A d$ where d is the submerged depth. Displaced a further x downwards, the resultant downward force is

$$m\ddot{x} = mg - \rho g A(d + x) = -\rho g A x.$$

Hence

$$\ddot{x} = -\frac{\rho g A}{m}\,x = -\frac{1025 \times 9.81 \times 0.030}{12}\,x = -25.14\,x \approx -25.1\,x.$$

So $\omega^2 = 25.14$, giving $\omega \approx 5.01\ \text{rad s}^{-1}$.
(b) Period: $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{5.01} \approx 1.25$ s. The motion starts from rest at maximum displacement, so $x = 0.20\cos(5.01\,t)$ and the amplitude is $a = 0.20$ m. Maximum speed:

$$v_{\max} = a\omega = 0.20 \times 5.01 \approx 1.00\ \text{m s}^{-1}.$$

(c) The buoy first passes through $x = 0$ when $\cos(5.01\,t) = 0$, i.e. $5.01\,t = \pi/2$, giving

$$t = \frac{\pi}{2\omega} = \frac{\pi}{10.03} \approx 0.31\,\text{s} \quad \left(= \tfrac{T}{4}\right).$$

(d) The model neglects the resistance of the water (viscous drag, roughly proportional to velocity). Including a term $-2mk\dot{x}$ gives damped harmonic motion, whose amplitude decays like e^{-kt} — matching the observed dying away. The SHM model is therefore only valid for the first few oscillations.

WARNING **GCE exam tip: use the stated result even if you cannot prove it**

Structured questions very often carry the instruction "show that $\ddot{x} = -\omega^2 x$" with a given numerical ω. If you cannot complete the proof, do not abandon the question: the later parts (period, amplitude, times, speeds) only need the stated value of ω. Write "from part (a), $\omega = \ldots$" and continue. You keep nearly all the marks.

4.9 Checking and Criticising a Model

A complete GCE answer often ends with a discussion mark. Here is what examiners expect.

Effect of neglected damping. If the statement says resistance is negligible but the real system loses energy, the SHM model overestimates the amplitude and the speeds after the first few cycles; the true period of a damped oscillator, $2\pi/\sqrt{\omega^2 - k^2}$, is slightly longer than $2\pi/\omega$.

Validity of the small-angle approximation. The pendulum equation is really $\ddot{\theta} = -\frac{g}{l}\sin\theta$; replacing $\sin\theta$ by θ is only accurate for small amplitudes (a few degrees). For swings of 30° or more, the true period is measurably longer than $2\pi\sqrt{l/g}$, and a g measurement made with large swings will be slightly too small.

Units and orders of magnitude. Before quoting a final answer, sanity-check it: a pendulum of length 1 m must have a period near 2 s; a car suspension should oscillate at roughly one cycle per second; a floating log bobs with a period of order one second. An answer of $T = 0.003$ s for a pendulum signals a units slip — almost always centimetres substituted as metres.

EXERCISE 6 **GCE practice**

A mass of 0.5 kg hangs from a light spring of natural length 0.6 m and modulus 30 N, extending it to equilibrium. The mass is pulled down a further 0.05 m and released. A resistance of magnitude $2v$ newtons acts, where v is the speed.

1. Show that $\ddot{x} + 4\dot{x} + 100x = 0$.
2. State the type of damping and write down the general solution.
3. Find the period of the damped oscillations and the time for the amplitude to fall to half

its initial value.

SOLUTION **Exercise 1**

1. Equilibrium: $mg = \lambda e/l \Rightarrow 0.5g = 30e/0.6$, so $e = 0.0981$ m. With a further displacement x downwards, tension $= \lambda(e + x)/l = 50(e + x)$, and

$$0.5\ddot{x} = 0.5g - 50(e + x) - 2\dot{x} = -50x - 2\dot{x},$$

using $0.5g = 50e$. Dividing by 0.5: $\ddot{x} + 4\dot{x} + 100x = 0$.
2. Here $k = 2$, $\omega = 10$, and $k^2 - \omega^2 = 4 - 100 < 0$: underdamped. General solution:

$$x = e^{-2t}\left(A\cos(\sqrt{96}\,t) + B\sin(\sqrt{96}\,t)\right), \qquad \sqrt{96} = 4\sqrt{6} \approx 9.80.$$

3. Damped period: $T = \dfrac{2\pi}{\sqrt{96}} \approx 0.641$ s. Amplitude decays as e^{-2t}; halving requires $e^{-2t} = \frac{1}{2}$, so $t = \dfrac{\ln 2}{2} \approx 0.347$ s.

KEY POINTS **Key points of Section 4**

- Follow the modelling checklist: equilibrium first, then $F = ma$ from equilibrium, then identify ω^2 (and k).
- A floating body of cross-section A in a liquid of density ρ performs SHM with $\omega^2 = \rho gA/m$.
- Engineers design suspensions for near-critical damping: fastest return to equilibrium without bouncing.
- Linearise $T^2 = 4\pi^2 l/g$ and take the gradient to measure g experimentally.
- Read the statement to choose between SHM and damped models; damping always enters with a negative sign.
- If you cannot prove the displayed equation, still use the stated ω in later parts.
- Always check units (SI throughout) and the order of magnitude of your final answer.

DID YOU KNOW? **Did you know?**

The same mathematics that keeps a car stable on the potholed roads between Bamenda and Douala also protects buildings during earthquakes: engineers fit giant tuned mass–

damper systems — effectively huge mass–spring–damper oscillators — inside skyscrapers so that the building's vibrations are absorbed and die out almost critically damped. The equation $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ quite literally saves lives.

5 INTEGRATION ACTIVITY

5.1 Aims of the activity

This integration activity closes the chapter by asking you to act as a **design engineer**. Working in groups of three or four, you will mobilise all the skills developed in this chapter:

- model a real mechanical system by a differential equation of the form $m\ddot{x} + \lambda\dot{x} + kx = 0$;
- compute the **natural angular frequency** and the undamped period of an oscillator;
- identify the three damping regimes (underdamped, critically damped, overdamped) from the discriminant of the auxiliary equation;
- solve a damped equation of motion with given initial conditions and sketch the response;
- compute the **damped period** and the **logarithmic decrement**, and interpret them physically;
- justify an engineering recommendation with rigorous mathematics, and defend it orally.

Beyond the calculations, the activity trains the GCE examiner's favourite skill: interpreting a mathematical result in the language of the real world.

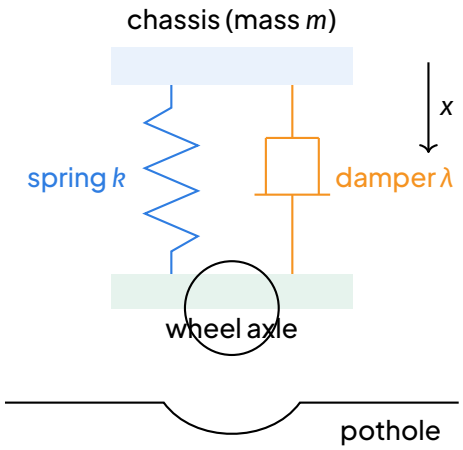
5.2 Situation

The road from Bamenda to Bafoussam, winding through the Western highlands, is famous for its beautiful scenery — and for its potholes. A transport company, Highland Express, runs pick-up trucks on this route and has received many complaints: passengers feel shaken long after each pothole, and goods arrive damaged. The company hires your group to advise on the **suspension** of its vehicles.

Each wheel of a pick-up truck carries an effective mass $m = 300$ kg (one quarter of the 1200 kg vehicle, neglecting the mass of the wheel itself). The suspension of each wheel is modelled as:

- a **spring** of stiffness $k = 20\,000$ N m^{-1}, and
- a **damper** (shock absorber) of damping coefficient λ (in N s m^{-1}),

mounted in parallel between the wheel axle and the chassis.



Quarter-car model of the suspension: spring and damper in parallel.

When the wheel hits a pothole, the chassis is suddenly displaced from its equilibrium position. We take the **equilibrium position** (with the truck at rest, the spring already compressed by the weight) as the origin, and we denote by $x(t)$ the vertical displacement of the chassis from equilibrium, measured downwards, in metres, at time t (in seconds) after the pothole impact.

Because the origin is the static equilibrium, the weight mg is exactly balanced by the static compression of the spring, so gravity does not appear in the equation of motion: only the extra spring force $-kx$ and the damping force $-\lambda\dot{x}$ act.

The company tells you that a typical pothole impact can be modelled as an **initial displacement** $x(0) = 0.05$ m (5 cm) with **zero initial velocity**, $\dot{x}(0) = 0$.

Your group must recommend a value of λ to the company, with full mathematical justification.

5.3 Tasks

Part A — Modelling and the undamped oscillator

1. Draw a force diagram for the chassis mass at a displacement x below equilibrium, and use Newton's second law to show that
$$m\ddot{x} + \lambda\dot{x} + kx = 0, \qquad \text{i.e.} \qquad 300\,\ddot{x} + \lambda\,\dot{x} + 20\,000\,x = 0.$$
2. Suppose first that the damper is removed ($\lambda = 0$). Show that the motion is simple harmonic, and compute the natural angular frequency ω and the undamped period T_0. Give T_0 to 3 decimal places. Comment: is this a comfortable frequency for passengers?

Part B — Critical damping

3. Write the auxiliary (characteristic) equation of the general equation of motion, and state the condition on λ for the damping to be critical.
4. Show that the critical damping coefficient is $\lambda_c = 2\sqrt{mk}$, and compute its numerical value.
5. Explain, using the form of the solutions in each regime, why car engineers choose damping close to critical: what happens to passengers if λ is too small? And if λ is too large?

Part C — The actual damper: $\lambda = 3000\ \text{N s m}^{-1}$

6. Compute the discriminant of the auxiliary equation and identify the damping regime.
7. Solve the equation of motion with $x(0) = 0.05$, $\dot{x}(0) = 0$. Write $x(t)$ in the form $x(t) = e^{-\alpha t}\left(A\cos\omega_d t + B\sin\omega_d t\right)$, giving α, ω_d, A, B to 3 significant figures.
8. Sketch the graph of $x(t)$ for $0 \le t \le 1.5$ s, marking the exponential envelope $\pm 0.05\,e^{-\alpha t}$.
9. Compute the damped period T_d and the logarithmic decrement $\delta = \ln\dfrac{x(t)}{x(t+T_d)} = \alpha T_d$.
10. Hence estimate how many visible oscillations the passengers feel after one pothole (an oscillation is "visible" while the amplitude exceeds about 1 mm). Is this acceptable comfort?

Part D — Recommendation

11. Write a short report (half a page) to Highland Express recommending a value (or range) of λ, justifying your choice mathematically (regime, settling time, number of oscillations) and in everyday language. Each group presents its recommendation to the class in three minutes.

5.4 Model answer

Part A. **1.** At displacement x below equilibrium, the spring is stretched by x beyond its static compression, so it pulls the chassis up with an extra force kx; the damper opposes the velocity with force $\lambda\dot{x}$. Newton's second law (positive downwards) gives

$$m\ddot{x} = -kx - \lambda\dot{x} \quad\Longrightarrow\quad 300\,\ddot{x} + \lambda\,\dot{x} + 20\,000\,x = 0.$$

2. With $\lambda = 0$: $\ddot{x} + \dfrac{20\,000}{300}x = 0$, i.e. $\ddot{x} + \omega^2 x = 0$ with

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{20\,000}{300}} = \sqrt{66.67} \approx 8.16\ \text{rad s}^{-1}, \qquad T_0 = \frac{2\pi}{\omega} \approx 0.770\ \text{s}.$$

This is SHM: without a damper the truck would bounce for ever about once every 0.77 s — extremely uncomfortable and dangerous.

Part B. **3.** The auxiliary equation is $300r^2 + \lambda r + 20\,000 = 0$. Critical damping corresponds to a repeated root: $\Delta = \lambda^2 - 4(300)(20\,000) = 0$.

4. $\lambda_c^2 = 4mk\cdot\ldots$ more precisely $\lambda_c = 2\sqrt{mk} = 2\sqrt{300 \times 20\,000} = 2\sqrt{6\,000\,000} \approx 4899\ \text{N s m}^{-1}$, i.e. $\lambda_c \approx 4900\ \text{N s m}^{-1}$.

5. If $\lambda < \lambda_c$ (underdamped), the solution contains a factor $\cos(\omega_d t - \varphi)$: the chassis oscillates through equilibrium several times before settling, so passengers feel repeated bouncing. If $\lambda > \lambda_c$ (overdamped), the return to equilibrium is a sum of two decaying exponentials, the slowest of which decays like $e^{r_2 t}$ with r_2 close to 0: the return is sluggish and the suspension feels "hard" over successive bumps. Critical damping gives the fastest return to equilibrium without oscillation, since $x = (A + Bt)e^{-\omega t}$ decays faster than any overdamped slow mode. Hence engineers aim for λ slightly below or equal to λ_c.

Part C. **6.** With $\lambda = 3000$: $\Delta = 3000^2 - 4(300)(20\,000) = 9\times10^6 - 2.4\times10^7 = -1.5\times10^7 < 0$: the motion is **underdamped**. The roots are

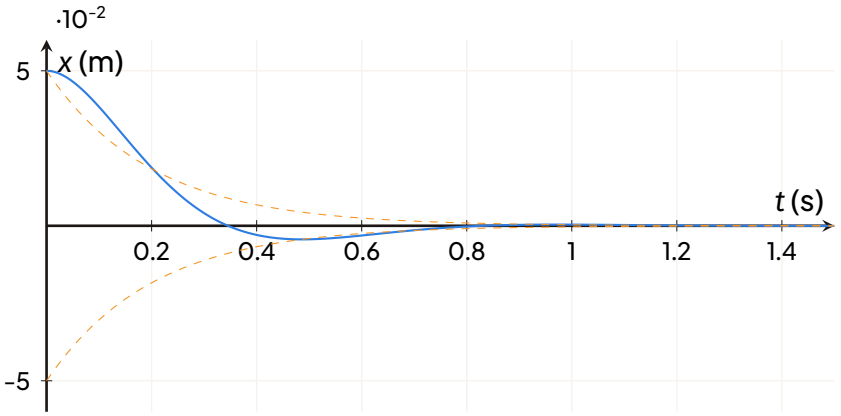
$$r = \frac{-3000 \pm \sqrt{-1.5\times10^7}}{600} = -5 \pm i\sqrt{\frac{1.5\times10^7}{600^2}} = -5 \pm i\,6.45.$$

So $\alpha = \dfrac{\lambda}{2m} = 5\ \text{s}^{-1}$ and $\omega_d = \sqrt{\omega^2 - \alpha^2} = \sqrt{66.67 - 25} = \sqrt{41.67} \approx 6.45\ \text{rad s}^{-1}$.

7. General solution: $x(t) = e^{-5t}\left(A\cos 6.455t + B\sin 6.455t\right)$. $x(0) = A = 0.05$. Differentiating, $\dot{x} = e^{-5t}\left[(-5A + 6.455B)\cos 6.455t + (-5B - 6.455A)\sin 6.455t\right]$, so $\dot{x}(0) = -5A + 6.455B = 0 \Rightarrow B = \dfrac{5(0.05)}{6.455} \approx 0.0387$.

$$\boxed{x(t) = e^{-5t}\left(0.0500\cos 6.45t + 0.0387\sin 6.45t\right)\text{ (m)}}$$

8. Sketch:



Underdamped response $x(t)$ with exponential envelope $\pm 0.05e^{-5t}$.

9. Damped period: $T_d = \dfrac{2\pi}{\omega_d} = \dfrac{2\pi}{6.455} \approx 0.974$ s. Logarithmic decrement:

$$\delta = \alpha T_d = 5 \times 0.974 \approx 4.87.$$

The ratio of successive amplitudes on the same side is $e^{-\delta} \approx e^{-4.87} \approx 0.0077$: after one full damped period the displacement is less than 1% of its initial value.

10. Amplitude after n periods: $0.05\,e^{-4.87n}$. For $n = 1$: $0.05 \times 0.0077 \approx 3.9\times10^{-4}$ m = 0.39 mm < 1 mm. So the passengers feel essentially **one single dip-and-recovery**, and no repeated bouncing. The motion settles within about one second. This is comfortable and acceptable.

Part D. **11.** Model report. We recommend a damper coefficient in the range $4000 \le \lambda \le 4900\ \text{N s m}^{-1}$. The present value $\lambda = 3000\ \text{N s m}^{-1}$ is only 61% of the critical value $\lambda_c \approx 4900$: although the oscillation dies out within one period, the first overshoot below equilibrium (about -0.036 m, read from the model) is large and the ride feels "soft and wallowing". Choosing λ close to λ_c eliminates the overshoot entirely, gives the fastest settling time $x = (A + Bt)e^{-8.16t}$, and protects both passengers and goods. Values above λ_c should be avoided: the slow exponential

mode would make the suspension react sluggishly to the rapid succession of potholes typical of the Bamenda–Bafoussam road. Recommendation: fit dampers of $\lambda \approx 4500\ \text{N s m}^{-1}$ per wheel, about 92% of critical damping.

WARNING **Marking points for the presentation**

Examiners (and your classmates) will check: correct use of the equilibrium origin (weight cancels the static spring force); correct auxiliary equation and discriminant; units carried throughout (rad s^{-1}, N s m^{-1}, s); and a conclusion that uses the numbers rather than vague phrases such as “the damping is good”.

6 METHODS AND WORKED EXAMPLES

6.1 Skill 1: Recognising and Setting Up the Equation of SHM

METHOD **Proving that a motion is simple harmonic**

To show that a particle moves with simple harmonic motion and to find its period:

1. Choose a **coordinate** x measured from the equilibrium position of the particle (this is essential: SHM is about oscillation about equilibrium).
2. Draw a diagram showing all forces acting at a general displaced position (weight, tension, spring force, etc.).
3. Apply Newton's second law $F = m\ddot{x}$ along the direction of motion, taking care with signs (use Hooke's law $T = \lambda e/l$ for springs or elastic strings).
4. Simplify until the equation takes the standard form
$$\ddot{x} = -\omega^2 x.$$
5. Read off ω^2, deduce ω, then the period $T = \dfrac{2\pi}{\omega}$ and the frequency $f = \dfrac{1}{T}$.

Reflex: if the equation comes out as $\ddot{x} = -\omega^2(x - a)$, the motion is still SHM, but about the shifted centre $x = a$; put $y = x - a$.

WORKED EXAMPLE 15 **Particle on a vertical spring**

A particle of mass 0.5 kg hangs in equilibrium from a light elastic spring of natural length 0.6 m and modulus of elasticity 12 N, the upper end being fixed. The particle is pulled down a further 0.1 m and released. Taking $g = 10\ \text{m s}^{-2}$, show that the particle moves with SHM and find the period of the motion.

SOLUTION

Step 1: equilibrium. Let e_0 be the extension at equilibrium. Then
$$\frac{\lambda e_0}{l} = mg \quad\Longrightarrow\quad \frac{12\,e_0}{0.6} = 0.5 \times 10 \quad\Longrightarrow\quad e_0 = 0.25\ \text{m}.$$

Step 2: equation of motion. Let x be the displacement below equilibrium. The total extension is $e_0 + x$, so the tension is $T = \dfrac{12}{0.6}(0.25 + x) = 5 + 20x$. Taking the downward direction as positive, Newton's second law gives
$$m\ddot{x} = mg - T = 5 - (5 + 20x) = -20x.$$

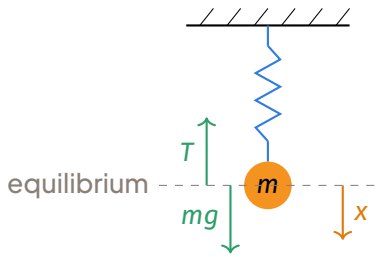
Hence

$$0.5\,\ddot{x} = -20x \quad \Longrightarrow \quad \ddot{x} = -40x.$$

This is of the form $\ddot{x} = -\omega^2 x$ with $\omega^2 = 40$, so the motion is simple harmonic about the equilibrium position. Step 3: period.

$$\omega = \sqrt{40} = 2\sqrt{10}, \qquad T = \frac{2\pi}{\omega} = \frac{2\pi}{2\sqrt{10}} = \frac{\pi}{\sqrt{10}} \approx 0.99\ \text{s}.$$

Remark: the amplitude is 0.1 m; since the spring remains extended throughout ($0.25 - 0.1 > 0$), the spring never goes slack and the SHM is complete.



Vertical spring: displacement x measured from the equilibrium position, with tension T upwards and weight mg downwards.

6.2 Skill 2: Using the Standard Solutions of SHM

METHOD **Exploiting $x = a\cos(\omega t + \varepsilon)$ and $v^2 = \omega^2(a^2 - x^2)$**

Given that a particle performs SHM with $\ddot{x} = -\omega^2 x$:

1. Identify the amplitude a (maximum displacement from the centre) and ω.
2. Write $x = a\cos(\omega t + \varepsilon)$, choosing ε from the initial conditions (if released from rest at $x = a$, simply $x = a\cos\omega t$).
3. Use the velocity formulas:
$$v = \dot{x} = -\omega a \sin(\omega t + \varepsilon), \qquad v^2 = \omega^2(a^2 - x^2).$$
4. Maximum speed: $v_{\max} = \omega a$ (at the centre); maximum acceleration: $\omega^2 a$ (at the extremes).
5. For a time between two positions, solve $x = a\cos(\omega t + \varepsilon)$ for t at each position and subtract.

Reflex: the formula $v^2 = \omega^2(a^2 - x^2)$ links speed and position without time — use it whenever time is not mentioned.

WORKED EXAMPLE 16 **Speeds and times in SHM**

A particle moves with SHM of period 2 s and amplitude 0.5 m. Find (a) the maximum speed; (b) the speed when the particle is 0.3 m from the centre; (c) the time taken to travel directly from $x = 0.3$ m to $x = -0.3$ m.

SOLUTION

We have $T = 2$ s, so $\omega = \dfrac{2\pi}{T} = \pi$ rad s^{-1}, and $a = 0.5$ m.

(a) Maximum speed:

$$v_{\max} = \omega a = \pi \times 0.5 = \frac{\pi}{2} \approx 1.57 \text{ m s}^{-1}.$$

(b) Speed at $x = 0.3$ m: using $v^2 = \omega^2(a^2 - x^2)$,

$$v^2 = \pi^2\left(0.5^2 - 0.3^2\right) = \pi^2(0.25 - 0.09) = 0.16\pi^2,$$

$$v = 0.4\pi \approx 1.26 \text{ m s}^{-1}.$$

(c) Time from $x = 0.3$ to $x = -0.3$: take $x = 0.5\cos(\pi t)$. At $x = 0.3$:

$$\cos(\pi t_1) = 0.6 \quad \Longrightarrow \quad \pi t_1 = \arccos(0.6) \approx 0.9273, \quad t_1 \approx 0.2952 \text{ s}.$$

At $x = -0.3$ (first time after passing the centre):

$$\cos(\pi t_2) = -0.6 \quad \Longrightarrow \quad \pi t_2 = \pi - 0.9273 = 2.2143, \quad t_2 \approx 0.7048 \text{ s}.$$

Hence the required time is

$$t_2 - t_1 \approx 0.7048 - 0.2952 = 0.410 \text{ s}.$$

Check: the particle passes the centre ($x = 0$) at $t = 0.5$ s, and indeed $0.2952 < 0.5 < 0.7048$. ■

6.3 Skill 3: Determining Amplitude and Phase from Initial Conditions

METHOD **Finding a and ε from $x(0)$ and $v(0)$**

1. Write the general solution $x = a\cos(\omega t + \varepsilon)$ (or equivalently $x = A\cos\omega t + B\sin\omega t$).
2. Substitute $t = 0$: $x(0) = a\cos\varepsilon$ and $v(0) = -\omega a\sin\varepsilon$.
3. Compute the amplitude from the energy-like identity

$$a^2 = x(0)^2 + \frac{v(0)^2}{\omega^2}.$$

4. Determine ε from the two equations of step 2 (signs of both sine and cosine fix the

quadrant).

Reflex: the form $x = A\cos\omega t + B\sin\omega t$ is often quicker: $A = x(0)$, $B = v(0)/\omega$.

WORKED EXAMPLE 17 **Motion started from an interior point**

A particle performs SHM with $\ddot{x} = -9x$ (units: m, s). At $t = 0$ the particle is at $x = 2$ m moving with speed $6\ \text{m s}^{-1}$ in the direction of x increasing. Find the amplitude, the maximum speed, and the first time at which the particle is instantaneously at rest.

SOLUTION

Here $\omega^2 = 9$, so $\omega = 3\ \text{rad s}^{-1}$.
Amplitude:

$$a^2 = x(0)^2 + \frac{v(0)^2}{\omega^2} = 2^2 + \frac{6^2}{3^2} = 4 + 4 = 8 \quad\Longrightarrow\quad a = 2\sqrt{2}\ \text{m}.$$

Maximum speed:

$$v_{\max} = \omega a = 3 \times 2\sqrt{2} = 6\sqrt{2} \approx 8.49\ \text{m s}^{-1}.$$

Phase and first rest position: write $x = 2\sqrt{2}\cos(3t + \varepsilon)$. At $t = 0$:

$$2\sqrt{2}\cos\varepsilon = 2 \;\Rightarrow\; \cos\varepsilon = \frac{1}{\sqrt{2}}, \qquad -3\cdot 2\sqrt{2}\sin\varepsilon = 6 \;\Rightarrow\; \sin\varepsilon = -\frac{1}{\sqrt{2}}.$$

Hence $\varepsilon = -\dfrac{\pi}{4}$, and $x = 2\sqrt{2}\cos\left(3t - \frac{\pi}{4}\right)$.
The particle is at rest at the extremes, i.e. when $3t - \frac{\pi}{4} = 0$ (first occasion, since it is moving towards $+x$):

$$t = \frac{\pi}{12} \approx 0.262\ \text{s}.$$

Check: at $t = \pi/12$, $x = 2\sqrt{2}\cos 0 = 2\sqrt{2} = a$. ■

6.4 Skill 4: Energy Methods in SHM

METHOD **Using kinetic and potential energy in SHM**

1. Recall that in SHM the total mechanical energy is constant:

$$E = \tfrac{1}{2}m\omega^2 a^2 = \tfrac{1}{2}mv_{\max}^2.$$

2. At displacement x: kinetic energy $E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(a^2 - x^2)$ and potential energy $E_p = \frac{1}{2}m\omega^2 x^2$.
3. To find where $E_k = E_p$ (or any given ratio), equate the expressions and solve for x.

4. For a spring, $E_p = \dfrac{\lambda x^2}{2l}$ measured from the natural length; energy conservation often replaces the equation of motion.

Reflex: "find the position where kinetic energy equals potential energy" $\Rightarrow a^2 - x^2 = x^2$, so $x = \pm\dfrac{a}{\sqrt{2}}$.

WORKED EXAMPLE 18 **Energy distribution in an oscillation**

A particle of mass 0.2 kg performs SHM with amplitude 0.4 m and period π s. (a) Find the total energy of the motion. (b) Find the kinetic energy when the particle is 0.2 m from the centre. (c) Find the positions where the kinetic energy is three times the potential energy.

SOLUTION

We have $\omega = \dfrac{2\pi}{T} = \dfrac{2\pi}{\pi} = 2\ \text{rad s}^{-1}$, $m = 0.2\ \text{kg}$, $a = 0.4\ \text{m}$.

(a) Total energy:

$$E = \tfrac{1}{2}m\omega^2 a^2 = \tfrac{1}{2}(0.2)(4)(0.16) = 0.064\ \text{J}.$$

(b) Kinetic energy at $x = 0.2$ m:

$$E_k = \tfrac{1}{2}m\omega^2(a^2 - x^2) = \tfrac{1}{2}(0.2)(4)(0.16 - 0.04) = 0.4 \times 0.12 = 0.048\ \text{J}.$$

(c) Positions where $E_k = 3E_p$:

$$\tfrac{1}{2}m\omega^2(a^2 - x^2) = 3 \times \tfrac{1}{2}m\omega^2 x^2 \quad\Longrightarrow\quad a^2 - x^2 = 3x^2 \quad\Longrightarrow\quad x^2 = \frac{a^2}{4}.$$

Hence

$$x = \pm\frac{a}{2} = \pm 0.2\ \text{m}.$$

Interpretation: at half the amplitude, the energy is shared in the ratio $3:1$ between kinetic and potential energy. ■

6.5 Skill 5: The Simple and Compound Pendulum

METHOD **Analysing pendulum motion**

1. For a simple pendulum of length l, write the tangential equation of motion for small angular displacement θ:

$$ml\ddot{\theta} = -mg\sin\theta \approx -mg\theta \quad\Longrightarrow\quad \ddot{\theta} = -\frac{g}{l}\theta, \qquad T = 2\pi\sqrt{\frac{l}{g}}.$$

2. For a compound pendulum (rigid body, axis through O), take moments about the axis:

$$I_O\ddot{\theta} = -mgh\sin\theta \approx -mgh\,\theta \quad\Longrightarrow\quad T = 2\pi\sqrt{\frac{I_O}{mgh}},$$

where h is the distance from O to the centre of mass and I_O the moment of inertia about the axis (use the parallel axes theorem $I_O = I_G + mh^2$).

3. The small-angle approximation $\sin\theta \approx \theta$ (with θ in radians) is what makes the motion simple harmonic — state it explicitly.

Reflex: in GCE questions, "show that the motion is approximately SHM" always signals the approximation $\sin\theta \approx \theta$.

WORKED EXAMPLE 19 **A rod swinging as a compound pendulum**

A uniform rod AB of length $2a$ and mass m is free to rotate in a vertical plane about a smooth horizontal axis through the end A. The rod is displaced through a small angle and released from rest. Given that the moment of inertia of the rod about its centre is $\frac{1}{3}ma^2$, show that the rod performs approximate SHM and find the period. Take $g = 9.8\ \text{m s}^{-2}$ and $a = 0.5$ m for the numerical value.

SOLUTION

Step 1: moment of inertia about A. By the parallel axes theorem, with $h = a$ (the centre of mass G is at the midpoint):

$$I_A = I_G + mh^2 = \tfrac{1}{3}ma^2 + ma^2 = \tfrac{4}{3}ma^2.$$

Step 2: equation of motion. Let θ be the angle the rod makes with the downward vertical. The weight mg acts at G, with perpendicular distance $a\sin\theta$ from the axis, giving a restoring moment $-mga\sin\theta$. Taking moments about A:

$$I_A\ddot{\theta} = -mga\sin\theta.$$

For small oscillations, $\sin\theta \approx \theta$, so

$$\tfrac{4}{3}ma^2\,\ddot{\theta} = -mga\,\theta \quad\Longrightarrow\quad \ddot{\theta} = -\frac{3g}{4a}\,\theta.$$

This is SHM with $\omega^2 = \dfrac{3g}{4a}$.

Step 3: period.

$$T = 2\pi\sqrt{\frac{4a}{3g}}.$$

With $a = 0.5$ m and $g = 9.8\ \text{m s}^{-2}$:

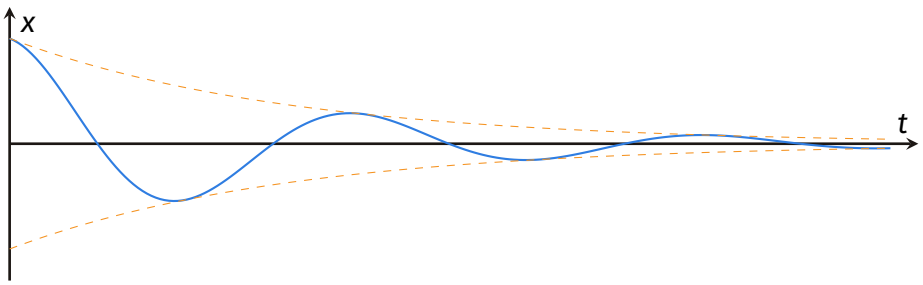
$$T = 2\pi\sqrt{\frac{4\times 0.5}{3\times 9.8}} = 2\pi\sqrt{\frac{2}{29.4}} \approx 2\pi\times 0.2609 \approx 1.64\ \text{s}.$$

Remark: this equals the period of a simple pendulum of length $\frac{4a}{3}$ (the "equivalent simple pendulum length"). ■

6.6 Skill 6: Solving Damped Harmonic Motion

METHOD **Analysing** $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$

1. Write the equation of motion including the resistance $-2mk\dot{x}$ (proportional to speed), and put it in the standard form
$$\ddot{x} + 2k\dot{x} + \omega^2 x = 0.$$
2. Write the auxiliary (characteristic) equation $\lambda^2 + 2k\lambda + \omega^2 = 0$, with roots $\lambda = -k \pm \sqrt{k^2 - \omega^2}$.
3. Classify the damping:
   - $k < \omega$ (light damping): $x = e^{-kt}(A\cos\mu t + B\sin\mu t)$ with $\mu = \sqrt{\omega^2 - k^2}$; oscillations with decaying amplitude, damped period $T_d = \frac{2\pi}{\mu}$.
   - $k = \omega$ (critical damping): $x = (A + Bt)e^{-kt}$; fastest return to equilibrium without oscillation.
   - $k > \omega$ (heavy damping): $x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$, both roots negative; slow return, no oscillation.
4. Use initial conditions to find A and B (remember to differentiate $x(t)$ before substituting $v(0)$).
5. The ratio of successive maximum displacements on the same side is e^{-kT_d} (constant); its logarithm kT_d is the logarithmic decrement.



Light damping: the displacement oscillates inside the decaying envelope $x = \pm ae^{-kt}$.

WORKED EXAMPLE 20 **Lightly damped oscillations**

A particle of mass 1 kg moves on a line under a restoring force $25x$ N towards the origin and a resistance of magnitude $4v$ N, where x is its displacement from O and v its speed. Initially $x = 2$ m and the particle is at rest. (a) Show that $\ddot{x} + 4\dot{x} + 25x = 0$. (b) Solve for x in terms of t. (c) Find the damped period and the ratio of two successive maximum displacements on the same side of O.

SOLUTION

(a) Equation of motion. Newton's second law with restoring force $-25x$ and resistance $-4\dot{x}$:

$$\ddot{x} = -25x - 4\dot{x} \quad \Longrightarrow \quad \ddot{x} + 4\dot{x} + 25x = 0.$$

(b) Solution. Standard form: $2k = 4$, so $k = 2$; $\omega^2 = 25$, so $\omega = 5$. Since $k = 2 < 5 = \omega$, the damping is light. The auxiliary equation is

$$\lambda^2 + 4\lambda + 25 = 0 \quad \Longrightarrow \quad \lambda = \frac{-4 \pm \sqrt{16 - 100}}{2} = -2 \pm \sqrt{21}\,i.$$

So $\mu = \sqrt{21}$ and

$$x = e^{-2t}\left(A\cos\sqrt{21}\,t + B\sin\sqrt{21}\,t\right).$$

Initial conditions: $x(0) = 2 \Rightarrow A = 2$. Differentiate:

$$\dot{x} = e^{-2t}\left(-\sqrt{21}A\sin\sqrt{21}t + \sqrt{21}B\cos\sqrt{21}t\right) - 2e^{-2t}\left(A\cos\sqrt{21}t + B\sin\sqrt{21}t\right),$$

$$\dot{x}(0) = \sqrt{21}\,B - 2A = 0 \quad \Longrightarrow \quad B = \frac{2A}{\sqrt{21}} = \frac{4}{\sqrt{21}}.$$

Hence

$$x = e^{-2t}\left(2\cos\sqrt{21}\,t + \frac{4}{\sqrt{21}}\sin\sqrt{21}\,t\right).$$

(c) Damped period and amplitude ratio.

$$T_d = \frac{2\pi}{\mu} = \frac{2\pi}{\sqrt{21}} \approx 1.371\ \text{s}.$$

Successive maxima on the same side are separated by one damped period, during which the envelope e^{-2t} is multiplied by

$$e^{-kT_d} = e^{-2\times 1.371} = e^{-2.742} \approx 0.0645.$$

So each maximum is about 6.4% of the previous one — the oscillations die out very quickly. ■

6.7 Skill 7: Critical and Heavy Damping — Applications

METHOD **Choosing and interpreting the damping regime**

1. Compute k and ω from the equation and compare them.
2. Critical damping ($k = \omega$) gives the quickest non-oscillatory return: this is the ideal for car suspensions, door closers and measuring instruments.
3. For $x = (A+Bt)e^{-kt}$, note that x can vanish at most once (when $t = -A/B$): the particle may cross the equilibrium position at most one time.
4. For heavy damping, both exponents are negative: $x \to 0$ without oscillation, dominated at large t by the slower exponent (the one closer to 0).

Reflex: exam questions asking "does the particle pass through O?" are answered by solving $x(t) = 0$ and checking whether the root is positive.

WORKED EXAMPLE 21 **Critically damped particle**

A particle of mass 0.5 kg moves under a restoring force and a resistance such that its equation of motion is $\ddot{x} + 8\dot{x} + 16x = 0$. Initially the particle is at O and is projected with speed 4 m s^{-1} away from O. (a) Solve for $x(t)$. (b) Find the maximum displacement from O and the time at which it occurs. (c) State, with justification, whether the particle ever returns through O.

SOLUTION

(a) Solution. Here $2k = 8$, so $k = 4$, and $\omega^2 = 16$, so $\omega = 4$. Since $k = \omega$, the damping is critical. The auxiliary equation $\lambda^2 + 8\lambda + 16 = 0$ has the repeated root $\lambda = -4$, so

$$x = (A + Bt)e^{-4t}.$$

Initial conditions: $x(0) = 0 \Rightarrow A = 0$. Then

$$\dot{x} = Be^{-4t} - 4Bt\,e^{-4t}, \qquad \dot{x}(0) = B = 4.$$

Hence

$$x = 4t\,e^{-4t}.$$

(b) Maximum displacement. Set $\dot{x} = 0$:

$$\dot{x} = 4e^{-4t}(1 - 4t) = 0 \quad \Longrightarrow \quad t = \frac{1}{4}\text{ s}.$$

Then

$$x_{\max} = 4 \times \frac{1}{4} \times e^{-1} = e^{-1} \approx 0.368\text{ m}.$$

(c) Return through O? We need $x(t) = 0$ for $t > 0$, i.e. $4t\,e^{-4t} = 0$. Since $e^{-4t} > 0$ always, this requires $t = 0$ only. Therefore the particle never passes through O again: it slows down, stops momentarily at $t = 0.25$ s, and creeps back towards O, reaching it only asymptotically as $t \to \infty$. ■

6.8 Skill 8: Composite GCE Problems — Broken and Combined Oscillations

METHOD **Handling interrupted or two-stage oscillations**

GCE questions often combine several skills: a string going slack, an elastic string becoming taut, a collision, or a mass detaching mid-oscillation.

1. Split the motion into phases and identify the governing law in each phase (SHM while the string is taut; free fall or uniform motion otherwise).
2. At each junction, compute the position and velocity from the previous phase — they are the initial conditions of the next.
3. Check the validity conditions: an elastic string only pulls (it must stay taut: extension ≥ 0); a spring can push and pull.
4. The total time of a "broken" oscillation is the sum of the times of the phases, found from $x = a\cos(\omega t + \varepsilon)$ in the SHM phases and from constant-acceleration formulas otherwise.

Reflex: always ask: "is the string still taut / is the contact maintained?" before applying SHM formulas over a whole swing. The string goes slack during the motion exactly when the amplitude exceeds the equilibrium extension e_0.

WORKED EXAMPLE 22 **Oscillation with a string going slack**

A particle P of mass 0.4 kg hangs in equilibrium from a light elastic string of natural length 0.8 m and modulus 8 N attached to a fixed point O. The particle is pulled down until the string has extension 1.0 m and is then released from rest. Take $g = 10\ \text{m s}^{-2}$. (a) Show that while the string is taut, P performs SHM and find its period. (b) Show that the string becomes slack during the motion. (c) Find the time from release until the string first becomes slack, and the speed of P at that instant.

SOLUTION

(a) SHM while taut. Equilibrium extension e_0: $\dfrac{8e_0}{0.8} = 0.4\times 10 \Rightarrow 10e_0 = 4 \Rightarrow e_0 = 0.4\,\text{m}$. Let x be the displacement below equilibrium; the tension is $T = \dfrac{8}{0.8}(0.4 + x) = 4 + 10x$. Newton's second law (downwards positive):

$$0.4\ddot{x} = 0.4 \times 10 - (4 + 10x) = -10x \quad \Longrightarrow \quad \ddot{x} = -25x.$$

SHM with $\omega = 5\ \text{rad s}^{-1}$, period $T = \dfrac{2\pi}{5} \approx 1.257\ \text{s}$.

(b) Does the string go slack? The particle is released from rest at extension 1.0 m, i.e. at $x = 1.0 - 0.4 = 0.6$ m below equilibrium, so the amplitude is $a = 0.6$ m. The SHM would carry the particle up to $x = -0.6$ m, i.e. to extension $0.4 - 0.6 = -0.2\ \text{m} < 0$, which is impossible

for a string. Since $a = 0.6 > e_0 = 0.4$, the string does go slack before the particle reaches the top of the would-be oscillation.
(c) Time and speed when slack. The string goes slack when the extension is 0, i.e. at $x = -0.4$ m. With release from rest at $x = 0.6$, write $x = 0.6\cos(5t)$:

$$0.6\cos(5t) = -0.4 \quad\Longrightarrow\quad \cos(5t) = -\frac{2}{3} \quad\Longrightarrow\quad 5t = \arccos\left(-\tfrac{2}{3}\right) \approx 2.3005.$$

$$t \approx \frac{2.3005}{5} \approx 0.460\ \text{s}.$$

The speed there follows from $v^2 = \omega^2(a^2 - x^2)$:

$$v = 5\sqrt{0.6^2 - 0.4^2} = 5\sqrt{0.36 - 0.16} = 5\sqrt{0.2} \approx 2.24\ \text{m s}^{-1} \quad \text{(upwards)}.$$

Thereafter the particle moves as a free projectile under gravity until the string tightens again — the starting point of the next phase. ■

WARNING **Frequent errors in the GCE**

- Measuring x from the top or from the natural length instead of from equilibrium: the equation then contains a constant term and candidates wrongly conclude the motion is not SHM. Shift the origin to equilibrium.
- Forgetting that $\sin\theta \approx \theta$ requires θ in radians.
- Confusing ω (angular frequency, rad s^{-1}) with f (frequency, Hz): $T = \dfrac{2\pi}{\omega} = \dfrac{1}{f}$.
- In damped motion, substituting $v(0)$ into $x(t)$ instead of into $\dot{x}(t)$.
- Applying SHM formulas after an elastic string has gone slack.

7 EXERCISES

7.1 I check my knowledge

EXERCISE 7 **Multiple choice**

A particle moves with simple harmonic motion according to $x = a\cos(\omega t)$. Its acceleration is

1. $\ddot{x} = \omega^2 x$, directed away from the centre;
2. $\ddot{x} = -\omega^2 x$, directed towards the centre;
3. $\ddot{x} = -\omega x$, directed towards the centre;
4. $\ddot{x} = \omega^2 a \sin(\omega t)$, directed away from the centre.

Choose the correct answer and justify your choice briefly.

EXERCISE 8 **True or false**

State whether each of the following statements is true or false, justifying your answer.

1. In SHM, the speed of the particle is greatest at the extreme positions.
2. The period of a simple pendulum is independent of the mass of the bob.
3. In a critically damped motion, the system returns to equilibrium without oscillating.
4. Light damping reduces the period of an oscillator compared with the undamped period.

EXERCISE 9 **Definitions**

Define each of the following terms as used in this chapter:

1. simple harmonic motion;
2. the amplitude of an oscillation;
3. critical damping;
4. the damped (quasi-)period of a lightly damped oscillator.

EXERCISE 10 **Direct calculations**

A particle performs SHM with $\ddot{x} = -25x$, where x is measured in metres.

1. Write down the angular frequency ω and the period T of the motion.
2. Given that the amplitude is 0.4 m, find the maximum speed and the maximum magnitude of the acceleration.
3. Find the speed of the particle when $x = 0.24$ m.

7.2 I practise

EXERCISE 11 **Mass on a horizontal spring**

A particle of mass 0.5 kg moves on a smooth horizontal table. It is attached to one end of a light elastic spring of natural length 0.8 m and modulus of elasticity 20 N, the other end of the spring being fixed to a point of the table. The particle is pulled 0.3 m beyond its equilibrium position and released from rest.

1. Show that the particle performs simple harmonic motion and find the angular frequency.
2. Find the period of the oscillations.
3. Find the maximum speed of the particle.
4. Find the time taken to reach, for the first time, a point at half the amplitude from the centre.

EXERCISE 12 **Vertical spring**

A mass hangs in equilibrium from a light vertical spring, extending it by 10 cm. The mass is pulled down a further 5 cm and then projected downward with speed 0.7 m s^{-1}. Take $g = 9.8$ m s^{-2}.

1. Prove that the subsequent motion is simple harmonic and find its angular frequency.
2. Find the amplitude of the motion.
3. Find the speed of the mass when it is 2 cm above the equilibrium position.

EXERCISE 13 **Timing in SHM**

A particle performs simple harmonic motion of period 4 s and amplitude 0.6 m.

1. Find the time it takes to travel directly from a point 0.3 m from the centre on one side to a point 0.3 m from the centre on the other side.
2. Find the speed of the particle at each of these two points.

EXERCISE 14 **Classifying damped motion**

The displacement x of a damped oscillator satisfies

$$\ddot{x} + 2k\dot{x} + 9x = 0,$$

where k is a positive constant.

1. Determine the values of k for which the motion is (a) oscillatory (light damping), (b) critically damped, (c) heavily damped.
2. In the critical case, given $x(0) = 0.2$ and $\dot{x}(0) = 0$, find $x(t)$ and show that the particle never crosses the equilibrium position.

7.3 I prepare for the GCE

EXERCISE 15 **The seconds pendulum — GCE style**

(12 marks)

A simple pendulum beats seconds (that is, its half-period is exactly 1 s) in Yaoundé, where $g = 9.78\ \text{m s}^{-2}$.

1. State the equation of motion of a simple pendulum for small oscillations and derive the formula for its period. (4 marks)
2. Calculate the length of this pendulum, giving your answer to four significant figures. (3 marks)
3. The pendulum is taken to a mountain station where its period increases by 0.2%. Find the new period and the value of g at the mountain station. (5 marks)

EXERCISE 16 **A damped oscillator — GCE style**

(14 marks)

A damped oscillator satisfies the equation

$$\ddot{x} + 4\dot{x} + 13x = 0,$$

with initial conditions $x(0) = 0.1$ and $\dot{x}(0) = 0$, where x is measured in metres and t in seconds.

1. Write down the auxiliary equation and solve it. (3 marks)
2. Hence find the general solution of the differential equation and apply the initial conditions to obtain $x(t)$ explicitly. (5 marks)
3. State the damped period of the oscillations. (2 marks)
4. Show that the ratio of two successive maximum displacements on the same side of the equilibrium position is constant, and find its value, giving your answer to three significant figures. (4 marks)

EXERCISE 17 **Floating log and car suspension — GCE style**

(16 marks)

Part A. A cylindrical log of mass $40\ \text{kg}$ and uniform cross-sectional area $0.05\ \text{m}^2$ floats vertically in water of density $1000\ \text{kg m}^{-3}$. The log is pushed down slightly from its equilibrium position and released. Take $g = 9.8\ \text{m s}^{-2}$.

1. Using Archimedes' principle, find the depth immersed when the log is in equilibrium. (3 marks)
2. Show that, neglecting resistance, the log performs simple harmonic motion, and find the period of the oscillations. (5 marks)
3. Discuss qualitatively how the resistance of the water would modify the motion, referring to the terms of a suitable differential equation. (3 marks)

Part B. A car's suspension is modelled by a mass of $250\ \text{kg}$, a spring of stiffness $18\,000\ \text{N m}^{-1}$ and a damper providing a resistance $\lambda\dot{x}$.

(d) The suspension must return to equilibrium as fast as possible without oscillating. Write down the equation of motion and find the required value of λ. (3 marks)

(e) If λ is only 60% of this value, classify the motion and find the damped period. (2 marks)

8 SOLUTIONS TO THE EXERCISES

9 WORKED EXERCISES: SIMPLE AND DAMPED HARMONIC MOTION

SOLUTION **Exercise 1 — Multiple Choice**

The correct answer is (b): $\ddot{x} = -\omega^2 x$, directed towards the centre.
Justification. Differentiating $x = a\cos(\omega t)$ twice:

$$\dot{x} = -a\omega \sin(\omega t), \qquad \ddot{x} = -a\omega^2 \cos(\omega t) = -\omega^2 x.$$

The acceleration is therefore proportional to the displacement and has the opposite sign: when $x > 0$ the acceleration is negative, and conversely. Hence it is always directed towards the centre O. This is precisely the defining property of simple harmonic motion.

SOLUTION **Exercise 2 — True or False**

1. False. From $v^2 = \omega^2(a^2 - x^2)$, the speed is greatest when $x = 0$, i.e. at the centre; at the extreme positions ($x = \pm a$) the speed is zero.
2. True. For a simple pendulum, $T = 2\pi\sqrt{\ell/g}$: the mass m cancels when the equation of motion $m\ell\ddot{\theta} = -mg\sin\theta$ is formed, so the period depends only on ℓ and g.
3. True. Critical damping ($k = \omega$ in $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$) gives $x = (A + Bt)e^{-kt}$, which has at most one zero; generically the system returns to equilibrium without oscillating, in the shortest possible time.
4. False. The damped angular frequency is $\omega_d = \sqrt{\omega^2 - k^2} < \omega$, so the damped period $T_d = 2\pi/\omega_d$ is longer than the undamped period $2\pi/\omega$. Light damping increases the period (slightly).

SOLUTION **Exercise 3 — Definitions**

1. Simple harmonic motion is the motion of a particle along a straight line whose acceleration is always directed towards a fixed point O of the line (the centre) and is proportional to its displacement from O: $\ddot{x} = -\omega^2 x$.

2. The amplitude is the maximum distance of the particle from the centre of the motion, i.e. half the distance between the two extreme positions.
3. Critical damping is the case of the equation $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$ in which $k = \omega$ (repeated auxiliary root); the system returns to equilibrium in the shortest time without oscillating.
4. The damped (quasi-)period of a lightly damped oscillator ($k < \omega$) is the time between two successive passages through the equilibrium position in the same direction: $T_d = \dfrac{2\pi}{\sqrt{\omega^2 - k^2}}$.

SOLUTION **Exercise 4 — Direct Calculations**

1. Comparing $\ddot{x} = -25x$ with $\ddot{x} = -\omega^2 x$: $\omega^2 = 25$, so $\omega = 5\ \text{rad s}^{-1}$ and
$$T = \frac{2\pi}{\omega} = \frac{2\pi}{5} \approx 1.26\ \text{s}.$$
2. With $a = 0.4\ \text{m}$:
$$v_{\max} = a\omega = 0.4 \times 5 = 2\ \text{m s}^{-1}, \qquad |\ddot{x}|_{\max} = \omega^2 a = 25 \times 0.4 = 10\ \text{m s}^{-2}.$$
3. Using $v^2 = \omega^2(a^2 - x^2)$ with $x = 0.24\ \text{m}$:
$$v^2 = 25\left(0.4^2 - 0.24^2\right) = 25(0.16 - 0.0576) = 25 \times 0.1024 = 2.56,$$
so $v = 1.6\ \text{m s}^{-1}$.

SOLUTION **Exercise 5 — Mass on a Horizontal Spring**

1. Let x be the displacement from the equilibrium position (which, on a smooth horizontal table, is the position where the spring has its natural length). The tension in the spring is $T = \dfrac{\lambda x}{l_0} = \dfrac{20}{0.8}x = 25x$, directed towards equilibrium. Newton's second law gives
$$0.5\,\ddot{x} = -25x \quad \Longrightarrow \quad \ddot{x} = -50x.$$
This is of the form $\ddot{x} = -\omega^2 x$ with $\omega^2 = 50$: the motion is SHM with $\omega = \sqrt{50} = 5\sqrt{2} \approx 7.07\ \text{rad s}^{-1}$.
2. $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{5\sqrt{2}} \approx 0.889\ \text{s}$.

3. The particle is released from rest at 0.3 m from the centre, so $a = 0.3$ m:

$$v_{\max} = a\omega = 0.3 \times 5\sqrt{2} = 1.5\sqrt{2} \approx 2.12 \text{ m s}^{-1}.$$

4. Measure time from release: $x = a\cos(\omega t) = 0.3\cos(5\sqrt{2}\,t)$. Half the amplitude: $x = 0.15$:

$$\cos(5\sqrt{2}\,t) = \tfrac{1}{2} \implies 5\sqrt{2}\,t = \frac{\pi}{3} \implies t = \frac{\pi}{15\sqrt{2}} \approx 0.148 \text{ s}.$$

SOLUTION **Exercise 6 — Vertical Spring**

1. At equilibrium, the spring is stretched by $e = 0.1$ m, so $mg = \dfrac{\lambda e}{l_0}$. Let x be the displacement measured downward from equilibrium. The total extension is $e+x$, so the upward tension is $\dfrac{\lambda(e+x)}{l_0}$. Newton's second law (downward positive):

$$m\ddot{x} = mg - \frac{\lambda(e+x)}{l_0} = -\frac{\lambda}{l_0}x,$$

since $mg = \lambda e/l_0$. Hence $\ddot{x} = -\dfrac{g}{e}x = -\dfrac{9.8}{0.1}x = -98x$: SHM with $\omega = \sqrt{98} = 7\sqrt{2} \approx 9.90 \text{ rad s}^{-1}$.

2. The mass starts at $x_0 = 0.05$ m with $\dot{x}_0 = 0.7 \text{ m s}^{-1}$. Using $v^2 = \omega^2(a^2 - x^2)$:

$$a^2 = x_0^2 + \frac{v_0^2}{\omega^2} = 0.05^2 + \frac{0.7^2}{98} = 0.0025 + 0.005 = 0.0075,$$

so $a = \sqrt{0.0075} \approx 0.0866 \text{ m} = 8.66 \text{ cm}$.

3. At $x = -0.02$ m (above equilibrium):

$$v^2 = 98\,(0.0075 - 0.0004) = 98 \times 0.0071 = 0.6958,$$

so $v \approx 0.834 \text{ m s}^{-1}$.

SOLUTION **Exercise 7 — Timing in SHM**

Here $T = 4$ s, so $\omega = \dfrac{2\pi}{4} = \dfrac{\pi}{2} \text{ rad s}^{-1}$, and $a = 0.6$ m.

1. Take $x = 0.6\cos\left(\frac{\pi}{2}t\right)$. The particle is at $x = +0.3$ when $\cos\left(\frac{\pi}{2}t\right) = \frac{1}{2}$, i.e. $\frac{\pi}{2}t_1 = \frac{\pi}{3}$, $t_1 = \frac{2}{3}$ s; it

reaches $x = -0.3$ when $\frac{\pi}{2}t_2 = \frac{2\pi}{3}$, $t_2 = \frac{4}{3}$ s. The direct journey takes

$$t_2 - t_1 = \frac{2}{3}\,\text{s} \approx 0.667\,\text{s}.$$

2. At $x = \pm 0.3$ m:

$$v^2 = \omega^2(a^2 - x^2) = \frac{\pi^2}{4}(0.36 - 0.09) = \frac{\pi^2}{4} \times 0.27, \qquad v = \frac{\pi}{2}\sqrt{0.27} \approx 0.817\ \text{m s}^{-1}.$$

The speed is the same at both points, as expected by symmetry.

SOLUTION **Exercise 8 — Classifying Damped Motion**

1. The auxiliary equation is $m^2 + 2km + 9 = 0$, with discriminant $\Delta = 4k^2 - 36 = 4(k^2 - 9)$.
   - (a) Oscillatory (light damping): $\Delta < 0 \iff 0 < k < 3$;
   - (b) Critical damping: $\Delta = 0 \iff k = 3$;
   - (c) Heavy damping: $\Delta > 0 \iff k > 3$.
2. For $k = 3$: $\ddot{x} + 6\dot{x} + 9x = 0$, auxiliary equation $(m + 3)^2 = 0$, so

$$x(t) = (A + Bt)e^{-3t}.$$

$x(0) = A = 0.2$; $\dot{x} = (B - 3A - 3Bt)e^{-3t}$, so $\dot{x}(0) = B - 3A = 0$, giving $B = 0.6$. Hence

$$x(t) = (0.2 + 0.6t)e^{-3t}.$$

Since $0.2 + 0.6t > 0$ for all $t \geq 0$ and $e^{-3t} > 0$, we have $x(t) > 0$ for all $t \geq 0$: the particle approaches the equilibrium position asymptotically but never crosses it.

SOLUTION **Exercise 9 — The Seconds Pendulum (GCE Style)**

Indicative mark scheme: (a) B1 equation of motion, M1 small-angle approximation, A1 SHM form, A1 period. (b) M1 rearranging, M1 substitution, A1 answer to 4 s.f. (c) B1 new period, M1 proportionality $T \propto 1/\sqrt{g}$, M1 substitution, A1 value of g, B1 units/comment.

1. For a simple pendulum of length ℓ, resolving along the tangent: $m\ell\ddot{\theta} = -mg\sin\theta$. For small oscillations, $\sin\theta \approx \theta$, giving

$$\ddot{\theta} = -\frac{g}{\ell}\theta,$$

which is SHM with $\omega^2 = g/\ell$, hence $T = 2\pi\sqrt{\ell/g}$.

2. A seconds pendulum has half-period 1 s, so $T = 2$ s:

$$\ell = \frac{gT^2}{4\pi^2} = \frac{9.78 \times 4}{4\pi^2} = \frac{9.78}{\pi^2} \approx 0.9910 \text{ m}.$$

3. New period: $T' = 2 \times 1.002 = 2.004$ s. Since $T = 2\pi\sqrt{\ell/g}$ with ℓ fixed, $g \propto 1/T^2$:

$$g' = g\left(\frac{T}{T'}\right)^2 = 9.78 \times \left(\frac{1}{1.002}\right)^2 \approx 9.78 \times 0.99602 \approx 9.741 \text{ m s}^{-2}.$$

(The decrease of g with altitude explains the increase of the period.)

Examiner's expectations: the small-angle approximation must be stated explicitly; in (c), using the ratio method avoids recomputing ℓ and is the expected elegant route.

SOLUTION **Exercise 10 — A Damped Oscillator (GCE Style)**

Indicative mark scheme: (a) B1 auxiliary equation, M1 solving, A1 complex roots. (b) B1 form of general solution, M1/M1 applying each initial condition, A1 constants, A1 final $x(t)$. (c) M1 use of ω_d, A1 period. (d) M1 envelope argument, M1 ratio over one period, A1 exponential value, A1 3 s.f.

1. Auxiliary equation: $m^2 + 4m + 13 = 0$;

$$m = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i.$$

2. Since the roots are complex ($-2 \pm 3i$), the motion is lightly damped and

$$x(t) = e^{-2t}(A\cos 3t + B\sin 3t).$$

$x(0) = A = 0.1$. Differentiating:

$$\dot{x} = e^{-2t}[(-2A + 3B)\cos 3t + (-3A - 2B)\sin 3t],$$

so $\dot{x}(0) = -2A + 3B = 0$, giving $B = \frac{0.2}{3} = \frac{1}{15} \approx 0.0667$. Hence

$$x(t) = e^{-2t}\left(0.1\cos 3t + \tfrac{1}{15}\sin 3t\right) \text{ (m)}.$$

3. Damped period: $T_d = \frac{2\pi}{3} \approx 2.09$ s.

4. Successive maxima on the same side are separated by one damped period T_d. Since the oscillatory factor repeats after T_d while the envelope is multiplied by e^{-2T_d}, the ratio of successive maximum displacements is the constant

$$\frac{x_{n+1}}{x_n} = e^{-2T_d} = e^{-4\pi/3} \approx 0.0152.$$

(Equivalently, the logarithmic decrement per cycle is $\frac{4\pi}{3} \approx 4.19$.)

Examiner's expectations: candidates must justify that maxima recur with period T_d (the cosine factor is periodic) and that the exponential envelope alone changes; quoting e^{-kT_d} without justification earns only partial credit.

SOLUTION **Exercise 11 — Floating Log and Car Suspension (GCE Style)**

Indicative mark scheme (Part A): (a) M1 Archimedes, M1 equilibrium equation, A1 depth. (b) M1 net force with extra immersion, A1 SHM equation, M1 identifying ω, M1 period formula, A1 value. (c) B1 damping term, B1 modified equation, B1 qualitative effect.
Part A.

1. At equilibrium, weight = upthrust. If d is the immersed depth, the displaced volume is $0.05d$, so
$$40g = 1000 \times 0.05\, d \times g \quad \Longrightarrow \quad d = \frac{40}{50} = 0.8\,\text{m}.$$

2. Let x be the additional downward displacement from equilibrium. The extra upthrust is $1000 \times 0.05\, x \times g = 490x$ (upward). Newton's second law:
$$40\ddot{x} = -490x \quad \Longrightarrow \quad \ddot{x} = -12.25x,$$
which is SHM with $\omega = \sqrt{12.25} = 3.5\,\text{rad s}^{-1}$. The period is
$$T = \frac{2\pi}{3.5} \approx 1.80\,\text{s}.$$

3. Water resistance opposes the velocity, adding a term $-\lambda\dot{x}$ to the resultant force:
$$40\ddot{x} + \lambda\dot{x} + 490x = 0.$$
The oscillations then decay exponentially in amplitude (light damping, quasi-period slightly increased), or the log settles without oscillating if the damping is critical or heavy.

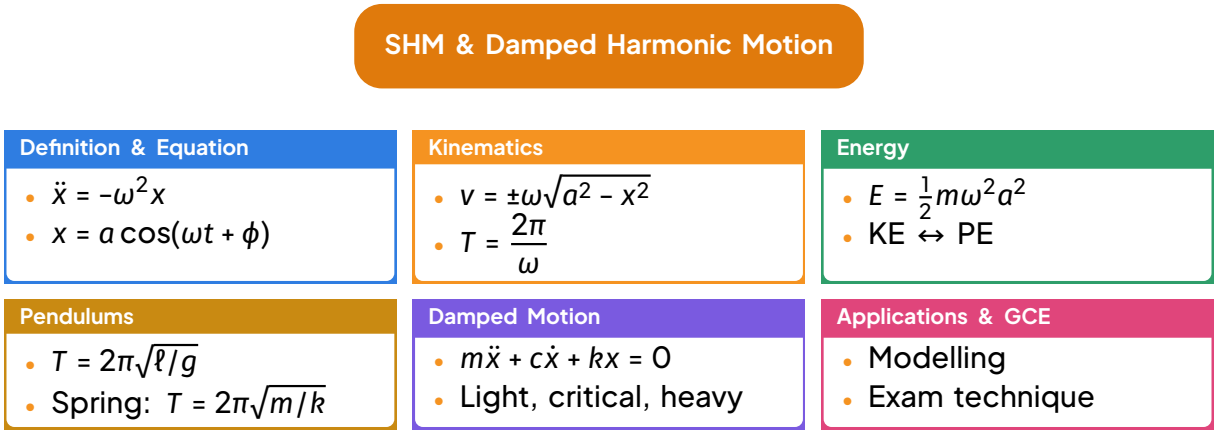
Part B. Indicative marks: (d) B1 equation, M1 critical condition, A1 value of λ; (e) B1 classification, A1 damped period.

(d) The equation of motion is
$$250\ddot{x} + \lambda\dot{x} + 18\,000x = 0.$$
Fastest return without oscillation requires critical damping: auxiliary equation $250m^2 + \lambda m + 18\,000 = 0$ with zero discriminant:
$$\lambda^2 = 4 \times 250 \times 18\,000 = 1.8 \times 10^7 \quad \Longrightarrow \quad \lambda = \sqrt{1.8 \times 10^7} \approx 4243\,\text{N s m}^{-1}.$$

(e) With $\lambda = 0.6 \times 4243 \approx 2546\,\text{N s m}^{-1}$, the discriminant is negative: the motion is lightly damped (oscillatory). Writing $\ddot{x} + 2k\dot{x} + \omega^2 x = 0$: $\omega^2 = \dfrac{18\,000}{250} = 72$ and $2k = \dfrac{2546}{250}$, so $k \approx 5.09$. Then
$$\omega_d = \sqrt{\omega^2 - k^2} = \sqrt{72 - 25.9} \approx 6.79\,\text{rad s}^{-1}, \qquad T_d = \frac{2\pi}{6.79} \approx 0.925\,\text{s}.$$

Examiner's expectations: in Part A the upthrust must be built from Archimedes' principle, not quoted; in Part B the critical-damping condition $\lambda = 2\sqrt{mk}$ (equivalently $\lambda^2 = 4 \times 250 \times 18\,000$) must be justified from the discriminant.

10 SUMMARY SHEET



Mind map of the chapter: Simple and Damped Harmonic Motion

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- **Simple harmonic motion** (SHM): motion about a fixed point O in which the acceleration is directed towards O and is proportional to the displacement from O: $\ddot{x} = -\omega^2 x$.
- **Amplitude** a: maximum displacement from the centre; **period** $T = \frac{2\pi}{\omega}$; **frequency** $f = \frac{1}{T}$; ω is the angular frequency.
- **Damped harmonic motion**: oscillatory motion with a resisting force proportional to velocity, $-c\dot{x}$, causing the amplitude to decay.
- **Light (under) damping**: oscillations with exponentially decaying amplitude; **critical damping**: fastest return to equilibrium without oscillation; **heavy (over) damping**: slow return without oscillation.

Formulas and results to know

- SHM equation: $\ddot{x} = -\omega^2 x$, with general solution $x = a\cos(\omega t+\phi)$, or $x = A\cos\omega t+B\sin\omega t$.
- Velocity: $\dot{x} = \pm\omega\sqrt{a^2 - x^2}$; maximum speed $v_{max} = \omega a$ at the centre; maximum acceleration $\omega^2 a$ at the extremes.
- Displacement from equilibrium: $x = a\cos\omega t$ (starting at extreme) or $x = a\sin\omega t$ (starting at centre).
- Energy: kinetic $\frac{1}{2}m\omega^2(a^2 - x^2)$, potential $\frac{1}{2}m\omega^2 x^2$, total $E = \frac{1}{2}m\omega^2 a^2$ (constant).

- Simple pendulum (small oscillations): $T = 2\pi\sqrt{\frac{\ell}{g}}$.
- Mass on a spring: $T = 2\pi\sqrt{\frac{m}{k}}$; for a vertical spring, oscillations are about the equilibrium position, and gravity only shifts the centre.
- Damping equation: $m\ddot{x} + c\dot{x} + kx = 0$; auxiliary equation $m\lambda^2 + c\lambda + k = 0$:
  - $c^2 < 4mk$: complex roots ⇒ light damping, $x = e^{-\alpha t}(A\cos\beta t + B\sin\beta t)$;
  - $c^2 = 4mk$: repeated root ⇒ critical damping, $x = (A + Bt)e^{-\alpha t}$;
  - $c^2 > 4mk$: real roots ⇒ heavy damping, $x = Ae^{\lambda_1 t} + Be^{\lambda_2 t}$.

Methods

- To prove SHM: choose a displacement x from equilibrium, apply $F = ma$, and reduce to $\ddot{x} = -\omega^2 x$; read off ω and T.
- To find amplitude and phase: use the initial conditions $x(0)$ and $\dot{x}(0)$.
- Time between two positions: solve $x(t) = x_0$ using the appropriate trig form, choosing the correct root from the direction of motion.
- For damped motion: form the auxiliary equation, classify the damping from the discriminant, then fit constants to the initial conditions.

Common mistakes

- Mixing up $x = a\cos\omega t$ and $x = a\sin\omega t$; always check against $x(0)$.
- Forgetting that $v = \pm\omega\sqrt{a^2 - x^2}$ has two signs; the direction of travel selects the sign.
- Using degrees instead of radians when evaluating $\omega t + \phi$.
- For a vertical spring, measuring x from the unstretched position instead of equilibrium, and wrongly concluding the motion is not SHM.
- Writing the damping force with the wrong sign, or classifying damping using c^2 against $4k$ instead of $4mk$.
- Confusing the damped angular frequency β with the natural frequency $\omega_0 = \sqrt{k/m}$.

GCE tips

- Examiners award a method mark for explicitly reaching the form $\ddot{x} = -\omega^2 x$ before quoting $T = \frac{2\pi}{\omega}$.
- State the centre of oscillation clearly; many marks are lost through a poorly defined variable.
- In "show that" questions, keep every step of the algebra; in numerical parts, give answers to 3 significant figures with units.

- For pendulum questions, justify the small-angle approximation $\sin\theta \approx \theta$ (in radians).
- Sketch displacement–time graphs for the three damping regimes: examiners reward correct shape, decay envelope $e^{-\alpha t}$, and correct intercepts.

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